16 Component Dirac Fermion Condensate Singlet Superfluid on 7+1 Deflating Spacetime

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A 16-component Dirac field is studied on a spacetime with seven space-like dimensions and one time-like dimension in which three of the space-like coordinates deflate exponentially as time advances. We derive the geometry, the spin connection and the Dirac equation, and solve the equation exactly. The deflation creates particles in every mode that depends on a deflating coordinate, with a Fermi-Dirac spectrum whose temperature is the rate divided by 2π. Fields that do not depend on the deflating coordinates obey the flat equations with couplings that grow with the deflation. The superfluid of fermion pairs bound by the Vorton-Scalar therefore acquires a history: its gap grows much faster than the coupling, two fermions become bound at a scattering resonance, and the paired state follows the deflation only once its gap exceeds a few times the rate divided by the pairing strength. With equal rates the spacetime is an exact vacuum solution of Einstein-Gauss-Bonnet gravity. The rotations that would make the pair condensate a Higgs field are no longer symmetries of the spacetime, and if the superfluid were the Higgs field, atomic clocks would limit the rate of deflation to a millionth of the Hubble rate.

I. INTRODUCTION

A. The question

Two earlier papers [1, 2] studied a single fermion field with sixteen components on two flat spacetimes of eight dimensions. One has four space-like and four time-like dimensions, and is written 4+4; the other has seven space-like dimensions and one time-like dimension, and is written 7+1. On both, the eight Dirac matrices can be chosen real, and the field supports a superfluid of fermion pairs whose order parameter points along one of the four coordinates that are not the coordinates of ordinary spacetime. This paper treats the 7+1 spacetime; a companion paper [3] treats 4+4.

The eight coordinates are written $x0, x1, \ldots, x7$. Throughout, $x1, x2, x3$ are the coordinates of ordinary space and $x4$ is the ordinary time. The coordinate $x0$ is an additional space-like coordinate, called the hidden coordinate. The remaining three, $x5, x6, x7$, are three additional *space-like* coordinates.

In the flat theory nothing explains why the fields of ordinary physics do not depend on $x5$, $x6$ and $x7$. On the 7+1 spacetime this was an assumption of the earlier paper. The present paper examines a definite proposal for what happens to these three coordinates.

B. The hypothesis of deflation

The hypothesis is that *the coordinates* $x5$*,* $x6$ *and* $x7$ *deflate exponentially*: as the ordinary time $x4$ advances, the proper extent of every interval along each of these coordinates shrinks as an exponential of $x4$. Flat spacetime is replaced by a curved spacetime with a diagonal metric, in which the three components of the metric along the deflating coordinates are

$$\begin{aligned} g_{55} &= -\mathrm{const5}^2\, e^{-2\,\mathrm{g55}\,x4}, \\ g_{66} &= -\mathrm{const6}^2\, e^{-2\,\mathrm{g66}\,x4}, \\ g_{77} &= -\mathrm{const7}^2\, e^{-2\,\mathrm{g77}\,x4}. \end{aligned} \tag{1}$$

Here g55, g66 and g77 are three positive constants, the *rates* of deflation; const5, const6 and const7 are three positive constants that set the scale at $x4 = 0$. The names of the rates are written upright and without a subscript, to distinguish them from the components g_{55}, g_{66}, g_{77} of the metric, which carry subscripts. The other five components of the metric keep their flat values. The minus signs in Eq. (1) are those of space-like components in the convention of this paper, in which the time-like component is positive.

The earlier papers examined three hypotheses about the superfluid, and we keep their labels.

H1. The fermion field forms a superfluid state of spin 0 in which the fermions are paired by the exchange of a new scalar boson, the *Vorton-Scalar*, through a neutral current of the kind known from the weak interaction.

H2. This superfluid state is the Higgs field.

H3. This superfluid state explains the observations of dark matter, or the Hubble tension, or both.

The purpose of this paper is to repeat every calculation of the earlier paper under the hypothesis of deflation, and to report what changes for H1, H2 and H3, favorable or not.

C. Summary of results

Geometry (Sec. II). The deflating spacetime is homogeneous and has 14 symmetries, or 15 or 20 if two or three

of the rates are equal, instead of the 36 of flat spacetime. It is not a solution of Einstein's equations without matter. With three equal rates it *is* an exact solution of the equations of Einstein-Gauss-Bonnet gravity without matter, for one definite relation between the rate and the two constants of that theory.

The Dirac field (Secs. III, IV). A field that does not depend on $x5, x6, x7$ obeys, after multiplication by the square root of the volume factor of the deflating coordinates, exactly the flat equation. A field that does depend on them is affected directly. We solve its equation exactly. *The deflation creates particles*: a mode that starts empty ends with the occupation of a Fermi-Dirac distribution at a temperature equal to the rate divided by 2π, the Gibbons-Hawking temperature of the de Sitter space that the deflating coordinates and the time form when the rates are equal.

The superfluid (Secs. V to X). For fields that do not depend on the extra coordinates the model of the earlier paper holds unchanged, with one difference: its coupling grows with the time, because a coupling of the four-dimensional description is a coupling of the eight-dimensional theory divided by a power of the deflating volume. The masses do not change. Every threshold of the earlier paper is therefore passed at a definite time. The gap of the paired state grows much faster than the coupling. Two fermions become bound at a scattering resonance. The paired state can follow the deflation only once its gap exceeds a few times the rate divided by the pairing strength; before that it lags behind and breaks pairs. Matter of a given density loses its fermion mass at a definite time. A vortex line carries four fermion zero modes.

Cosmology (Sec. XI). If gravity is the Einstein theory in eight dimensions, Newton's constant grows at the total rate of deflation; lunar laser ranging then bounds that rate to 3×10^{-3} of the Hubble rate, and the Hubble constant inferred from the microwave background falls. The deflation does not relieve the Hubble tension.

The Higgs hypothesis (Sec. XII). The rotations of the four extra space-like coordinates, under which the pair condensate has the pattern of the Higgs vacuum, are no longer symmetries of the spacetime; the pattern survives only as a hidden symmetry. If the condensate were the Higgs field, the electroweak scale would follow the growing coupling, and atomic clocks would limit the rate of deflation to about 10^{-6} of the Hubble rate. The hypothesis H2 and a deflation of any consequence today exclude each other.

D. Method, and the limits of what is claimed

Every statement of this paper about geometry, matrices and operators is an identity that a program has verified in exact rational arithmetic, and that an independent program has verified again by computer algebra for arbitrary values of the rates. Every numerical statement is the output of a program, with a stated accuracy. The programs, their output and the documents that derive each result step by step are public [4]; the program performs 217 checks, all of which pass. The metric of Eq. (1) is a hypothesis. The paper derives its consequences; it does not claim that the hypothesis is true. Section XIII lists what has been established and what has not.

Units are natural, $\hbar = c = k_B = 1$. Symbols that carry the tag $1671d$ refer to the deflating 7+1 spacetime. A general coordinate among the three deflating ones is written xt with $t \in \{5, 6, 7\}$ and its rate rate_t; a general coordinate among $x0, x1, x2, x3$ is written xj with $j \in \{0, 1, 2, 3\}$. The letter a is kept for the four extra axes, $a \in \{0, 5, 6, 7\}$ (Sec. VI). Appendix A lists the symbols.

II. THE DEFLATING SPACETIME

This section derives the geometry that the hypothesis implies: the curvature, the field equations that the metric does and does not solve, its symmetries, and the motion of a classical particle. The derivations are given in full in Ref. [4].

A. Metric, frame and volume

Let $\eta = \text{diag}(-1, -1, -1, -1, +1, -1, -1, -1)$ be the flat metric of the earlier paper, with the coordinates in the order $x0, \ldots, x7$. Define three *scale factors*, one for each deflating coordinate,

$$b_t(x4) = \text{const}_t \, e^{-\text{rate}_t \, x4}, \qquad t = 5, 6, 7, \tag{2}$$

with $\text{rate}_5 = \text{g55}$, $\text{rate}_6 = \text{g66}$, $\text{rate}_7 = \text{g77}$. The metric of the deflating spacetime is

$$g_{\mu\nu} = \text{diag}\big(\eta_{00}, \eta_{11}, \eta_{22}, \eta_{33}, \eta_{44}, \, \eta_{55} b_5^2, \, \eta_{66} b_6^2, \, \eta_{77} b_7^2\big), \tag{3}$$

whose last three components are those of Eq. (1). Greek indices label the coordinates. A coordinate interval dxt has the proper extent $b_t \, |dxt|$, which shrinks by the factor e in the time $1/\text{rate}_t$. Because $x5, x6, x7$ are space-like, what shrinks is a length.

Capital indices A, B label an orthonormal frame, eight vector fields $e_A = e_A{}^\mu \partial_\mu$ with $g_{\mu\nu} e_A{}^\mu e_B{}^\nu = \eta_{AB}$. We take the frame along the coordinates, $e_A{}^\mu = \text{diag}(1, 1, 1, 1, 1, b_5^{-1}, b_6^{-1}, b_7^{-1})$. The volume element is $V_d \, d^8x$ with

$$V_d(x4) = b_5 b_6 b_7 \propto e^{-\text{gsum}\, x4}, \quad \text{gsum} = \text{g55} + \text{g66} + \text{g77}. \tag{4}$$

V_d is the proper volume of a unit coordinate cell of the deflating coordinates, and the *total rate* gsum is the rate at which it shrinks. Most results below depend on the three rates only through gsum.

B. Curvature

The only components of the metric that depend on a coordinate are the three g_{tt}, and they depend on $x4$ alone. The Christoffel symbols that do not vanish are

$$\Gamma^t{}_{4t} = -\mathrm{rate}_t, \qquad \Gamma^4{}_{tt} = \eta_{44}\eta_{tt}\,\mathrm{rate}_t\, b_t^2, \tag{5}$$

without summation over t. The Riemann tensor has, in the orthonormal frame, the components

$$R^t{}_{4t4} = -\mathrm{rate}_t^2, \quad R^t{}_{sts} = -\eta_{44}\eta_{ss}\,\mathrm{rate}_t\,\mathrm{rate}_s \quad (t \neq s), \tag{6}$$

and those related to them by symmetry. No component carries an index of $x0, \ldots, x3$: these four coordinates span a flat space, and the curvature lives in the four coordinates $x4, \ldots, x7$. With $\mathrm{gsq} = \mathrm{g55}^2 + \mathrm{g66}^2 + \mathrm{g77}^2$ and $\mathrm{gprod} = \mathrm{g55g66} + \mathrm{g55g77} + \mathrm{g66g77}$, the curvature scalar is the constant $R = -\eta_{44}(\mathrm{gsq} + \mathrm{gsum}^2)$, and the Einstein tensor $\mathcal{G}^\mu{}_\nu = R^\mu{}_\nu - \frac{1}{2}R\,\delta^\mu_\nu$ is diagonal with

$$\begin{aligned} \mathcal{G}^4{}_4 &= \eta_{44}\,\mathrm{gprod}, \qquad \mathcal{G}^j{}_j = \tfrac{1}{2}\eta_{44}\,(\mathrm{gsq} + \mathrm{gsum}^2), \\ \mathcal{G}^t{}_t &= \eta_{44}\big[\tfrac{1}{2}(\mathrm{gsq} + \mathrm{gsum}^2) - \mathrm{rate}_t\,\mathrm{gsum}\big]. \end{aligned} \tag{7}$$

The Einstein tensor does not vanish, and it is not a multiple of the unit matrix unless all rates vanish. *The deflating spacetime is not a solution of Einstein's equations without matter, with or without a cosmological constant.*

C. A solution of Einstein-Gauss-Bonnet gravity

In more than four dimensions the general field equations of second order for the metric are those of Lovelock [5]. Their first term beyond Einstein's is the Lanczos tensor $\mathcal{H}^\mu{}_\nu$ of the Gauss-Bonnet term [6], and the equations without matter read

$$\mathcal{G}^\mu{}_\nu + \Lambda_8\,\delta^\mu_\nu + \alpha_{\mathrm{GB}}\,\mathcal{H}^\mu{}_\nu = 0, \tag{8}$$

with a cosmological constant Λ_8 and a coupling α_{GB}. (In this subsection the signs are those of the convention in which space-like components of the metric are positive.) For the metric (3) we find

$$\begin{aligned} \mathcal{H}^j{}_j &= -4\,\mathrm{g55}\,\mathrm{g66}\,\mathrm{g77}\,\mathrm{gsum} \quad (j = 0, 1, 2, 3), \\ \mathcal{H}^4{}_4 &= \mathcal{H}^t{}_t = 0, \end{aligned} \tag{9}$$

and the Lovelock tensor of the third order, the last that can differ from zero in eight dimensions, vanishes identically. The components of Eq. (8) along $x4, \ldots, x7$ therefore contain no Gauss-Bonnet term and require the four corresponding components of the Einstein tensor to be equal, which holds if and only if the three rates are equal. The components along $x0, \ldots, x3$ then fix the coupling. The result:

$$\mathrm{g55} = \mathrm{g66} = \mathrm{g77}: \qquad \Lambda_8 = 3\,\mathrm{g55}^2, \qquad \alpha_{\mathrm{GB}} = -\frac{1}{4\,\mathrm{g55}^2}. \tag{10}$$

With three equal rates the deflating spacetime is an exact solution of Einstein-Gauss-Bonnet gravity without matter, in a theory whose two constants satisfy $4\alpha_{\mathrm{GB}}\Lambda_8 = -3$; the rate is then $\mathrm{g55} = \sqrt{\Lambda_8/3}$. With unequal rates it is not a solution for any Λ_8 and α_{GB}. Solutions of this theory in which every scale factor is an exponential of the time have been classified in five and six dimensions [7]; the present one, with four static and three contracting directions in eight dimensions, is of that kind. Whether it is stable has not been examined.

D. Symmetries

A Killing vector is a vector field along which the metric does not change. Flat spacetime in eight dimensions has 36. The deflating spacetime has, for any rates, 14: the seven translations along all coordinates but $x4$; the six rotations of $x0, x1, x2, x3$; and the scaling

$$\mathcal{X}_{\mathrm{scale}} = \partial_4 + \mathrm{g55}\,x5\,\partial_5 + \mathrm{g66}\,x6\,\partial_6 + \mathrm{g77}\,x7\,\partial_7, \tag{11}$$

which shifts the time and stretches the deflating coordinates so as to undo the shrinking. Two equal rates add the rotation of the two coordinates concerned. Three equal rates add the three rotations of $x5, x6, x7$ and three further vectors, 20 in all; the four coordinates $x4, \ldots, x7$ then form a space of constant curvature of radius $1/\mathrm{g55}$: de Sitter space in the half of its flat slicing in which space contracts. These lists are complete. (A Killing vector that vanishes at a point is a Lorentz transformation of the tangent space that leaves the Riemann tensor (6) unchanged; the dimension of the space of such transformations is 6, 7 or 12, which bounds the number of Killing vectors by 14, 15 or 20.)

Three consequences are used below. The spacetime is homogeneous: no place and no moment is distinguished. The constants const_t are not observable by themselves, since a rescaling of xt changes them; the deflation is observable only through an object that fixes a coordinate scale along xt, such as a wave of given coordinate wavelength. And there is no conserved energy, because a translation in time alone is not a symmetry.

Of the 28 rotations and boosts of flat spacetime, only the six rotations of $x0, \ldots, x3$ survive for general rates (Fig. 1). Nothing that involves the time survives, in particular not the boosts of ordinary special relativity: the deflation defines a preferred state of rest, as the expansion of the universe does.

E. Geodesics

The translations give seven conserved momenta P_μ ($\mu \neq 4$) of a classical particle. Let $E_\perp = (m^2 + \sum_j P_j^2)^{1/2}$, with the sum over $j = 0, 1, 2, 3$, be the energy that a particle of mass m would have without momentum along the deflating coordinates. The conservation of the

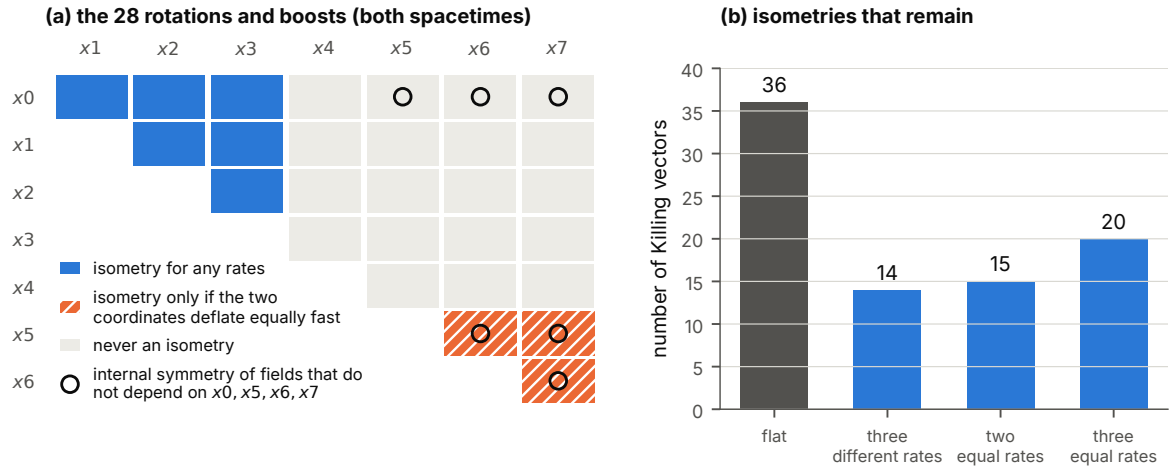


FIG. 1. Symmetries of the deflating spacetime. (a) The 28 rotations and boosts of flat spacetime, one square for each pair of coordinates: those that remain symmetries for any rates, those that remain symmetries only if the two coordinates deflate equally fast, and those that are never symmetries. Circles mark the six pairs whose rotations act as internal symmetries on fields that do not depend on $x0, x5, x6, x7$ (Sec. XII). (b) The number of Killing vectors.

norm of the momentum gives the velocity along the time, with an affine parameter ς along the path:

$$\Big(\frac{dx4}{d\varsigma}\Big)^2 = E_\perp^2 + \sum_t \frac{P_t^2}{b_t(x4)^2}. \tag{12}$$

The last term grows as $e^{2\,\mathrm{rate}_t\,x4}$: the momentum P_t belongs to a fixed coordinate wavelength, and the proper wavelength shrinks. With the plus sign the energy of a particle that moves along a deflating coordinate grows without bound, the reverse of the redshift in an expanding universe. For momentum along $x5$ alone the proper time that elapses between $x4$ and $x4 = \infty$ is

$$\frac{m}{\mathrm{g55}\,E_\perp}\,\mathrm{arsinh}\Big(\frac{E_\perp\,\mathrm{const5}\,e^{-\mathrm{g55}\,x4}}{|P_5|}\Big), \tag{13}$$

which is finite [Fig. 2(a)]: the coordinates of Eq. (3) do not cover the whole future of such a particle. For equal rates this is the known incompleteness of the flat slicing of de Sitter space. Section IV shows the quantum form of the blueshift. (On the 4+4 spacetime the sign is the opposite, and such a particle is turned back [3].)

III. THE DIRAC FIELD

A. Matrices and spin connection

At each point of a curved spacetime the tangent space is flat. The Dirac matrices of the orthonormal frame are therefore the eight real 16×16 matrices γ^A of the earlier paper [2], which satisfy $\gamma^A\gamma^B + \gamma^B\gamma^A = 2\eta^{AB}$; all of their properties were verified again for this work. The matrices with a coordinate index are $\gamma^\mu = e_A{}^\mu\gamma^A$: five of them are the constant matrices, and the three along the deflating coordinates are the constant matrices divided by the scale factors, $\gamma^{A=t}/b_t$. They satisfy $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$.

A spinor is defined relative to a frame, and its derivative must account for the turning of the frame from point to point [8]. The covariant derivative of a spinor is $\nabla_\mu\psi = (\partial_\mu + \Omega_\mu)\psi$ with the spin connection $\Omega_\mu = \frac{1}{4}\omega_{\mu AB}\gamma^A\gamma^B$, whose coefficients follow from the metric. For the metric (3) the only coefficients that do not vanish are $\omega_{t\,t4} = -\omega_{t\,4t} = -\eta_{tt}\,\mathrm{rate}_t\,b_t$, so that

$$\begin{aligned} &\Omega_t = -\tfrac{1}{2}\,\eta_{tt}\,\mathrm{rate}_t\,b_t\,\gamma^t\gamma^4 \quad (t = 5, 6, 7),\\ &\Omega_0 = \cdots = \Omega_4 = 0, \end{aligned} \tag{14}$$

with the constant matrices of the frame on the right. Each Ω_t generates the Lorentz transformation in the plane of xt and $x4$. The spin connection is correct if and only if the Dirac matrices are covariantly constant, $\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0$, which holds for all 64 pairs of indices. Its curvature is the Riemann tensor, $[\nabla_\mu, \nabla_\nu]\psi = \frac{1}{4}R_{AB\mu\nu}\gamma^A\gamma^B\psi$, a third route to Eq. (6). One combination enters the Dirac equation:

$$\gamma^\mu\Omega_\mu = -\tfrac{1}{2}\,\mathrm{gsum}\,\gamma^4. \tag{15}$$

The scale factors cancel, and only the total rate remains.

B. The Dirac equation

The field of this paper, $\psi_{1671d}(x)$, is a column of sixteen complex-valued Grassmann operators. Its equation is the equation of the earlier paper with the covariant derivative and the matrices γ^μ:

$$\big(i\gamma^\mu\nabla_\mu - m\big)\psi_{1671d} = 0. \tag{16}$$

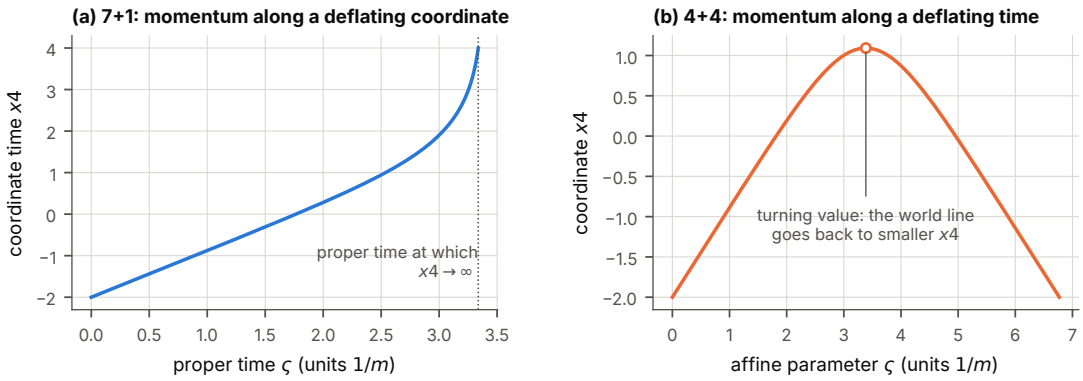


FIG. 2. A classical particle of the mass $m = 1$ with the momenta $P_1 = 0.5$ along $x1$ and $P_5 = 0.3$ along the deflating coordinate $x5$, for g55 = 1 and const5 = 0.8, started at $x4 = -2$. (a) On the 7+1 spacetime the time $x4$ grows without bound within the finite proper time 3.386. (b) On the 4+4 spacetime the path reaches the turning value $x4 = 1.0924$ and returns to smaller $x4$.

With Eq. (15),

$$i\gamma^4\Big(\partial_4 - \frac{\text{gsum}}{2}\Big)\psi + i\sum_{j=0}^{3}\gamma^j\partial_j\psi + i\sum_{t=5}^{7}\frac{\gamma^t}{b_t}\,\partial_t\psi - m\,\psi = 0, \tag{17}$$

with the constant matrices. Two things have changed with respect to flat spacetime: the derivative along the time is shifted by gsum/2, and each derivative along a deflating coordinate is divided by its scale factor.

C. The weighted field

The shift is removed by a change of normalization. Define the *weighted field*

$$\hat{\psi} = \sqrt{V_d}\,\psi. \tag{18}$$

Because $\partial_4\sqrt{V_d} = -\frac{1}{2}\text{gsum}\sqrt{V_d}$, the weighted field obeys Eq. (17) without the term gsum/2. The quantity $\psi^\dagger\psi$ is a density per unit of proper volume and grows as the deflating coordinates shrink; $\hat{\psi}^\dagger\hat{\psi}$ is the density per unit of coordinate volume, from which this trivial growth is removed.

The central observation of this paper follows at once. *For a field that does not depend on $x5$, $x6$, $x7$, the equation of the weighted field is exactly the flat equation.* Every result of the earlier paper for such fields, in particular everything about the pairing of the fermions, holds for the weighted field. The deflation enters those results in one way only, through the dependence of the couplings on V_d (Sec. V).

A field that does depend on a deflating coordinate feels the deflation directly. Because the coefficients depend on $x4$ only, a solution can be taken proportional to $\exp(i\sum_j p_j\,xj + i\sum_t k_t\,xt)$ with seven constant wavenumbers, four along the flat coordinates xj, $j = 0, 1, 2, 3$, and three along the deflating ones; such a solution is a *mode*. Multiplying the equation of the weighted field by γ^4 gives an ordinary differential equation in the time:

$$i\,\partial_4\hat{\psi} = \mathsf{h}(x4)\,\hat{\psi}, \qquad \mathsf{h} = \mathsf{h}_\perp + k_d(x4)\,\alpha^d, \tag{19}$$

where $\mathsf{h}_\perp = m\beta + \sum_j p_j\alpha^j$ is the Hamiltonian of the modes that do not depend on the deflating coordinates ($\beta = \gamma^4$, $\alpha^A = \gamma^4\gamma^A$),

$$k_d(x4) = \Big(\sum_t \frac{k_t^2}{b_t(x4)^2}\Big)^{1/2} \tag{20}$$

is the *physical* wavenumber along the deflating coordinates, and $\alpha^d = k_d^{-1}\sum_t(k_t/b_t)\alpha^t$ is the matrix α along its direction. With $E_\perp = (m^2 + \sum_j p_j^2)^{1/2}$,

$$\mathsf{h}^2 = \big(E_\perp^2 + k_d^2\big)\,1. \tag{21}$$

The Hamiltonian is real and symmetric at every time: the evolution is unitary and $\hat{\psi}^\dagger\hat{\psi}$ is conserved. But the Hamiltonian depends on the time, through a term that anticommutes with β, and the energy of a mode with $k_t \neq 0$ grows as $e^{\text{rate}_t\,x4}$. A Hamiltonian of this kind creates pairs [9].

IV. MODES THAT DEPEND ON THE DEFLATING COORDINATES

A. Three kinds of modes

Modes fall into three kinds. A mode that does not depend on the deflating coordinates, $k_5 = k_6 = k_7 = 0$, obeys the flat equation; its solutions are the plane waves of the earlier paper, with the energies $\pm E_\perp$, eight of each sign. A mode with some $k_t \neq 0$ behaves in the same way as long as its physical wavenumber is negligible, $k_d \ll E_\perp$. But k_d grows without bound, and every such mode sooner or later reaches $k_d \gg E_\perp$. The moment at which $k_d = E_\perp$ is its *crossing*. The crossing is where

the deflation acts on the field, and this section solves the equation of the weighted field across it.

The solution is exact when *one rate governs*: when only one of the k_t differs from zero, or when the three rates are equal. In both cases the matrix α^d does not depend on the time, and k_d is proportional to $e^{\mathrm{rate}_t\, x4}$ with the rate that governs. The natural variable is then the physical wavenumber in units of the rate,

$$u_d = \frac{k_d(x4)}{\mathrm{rate}_t}, \qquad \frac{d}{dx4} = \mathrm{rate}_t\, u_d\, \frac{d}{du_d}. \tag{22}$$

The crossing is at $u_d = E_\perp/\mathrm{rate}_t$, and this ratio of the energy to the rate is the one parameter on which the results depend.

B. The exact solution

In $\mathsf{h} = \mathsf{h}_\perp + k_d\,\alpha^d$ the two constant matrices satisfy $\mathsf{h}_\perp^2 = E_\perp^2$ and $(\alpha^d)^2 = 1$, and they anticommute. Let w be one of the eight eigenvectors of $\mathsf{h}_\perp$ with the eigenvalue $+E_\perp$; then $\alpha^d w$ is an eigenvector with the eigenvalue $-E_\perp$. Let Z_+ and Z_- denote two cylinder functions of the same kind (Bessel functions of the first or of the second kind, or Hankel functions) of the orders $\frac{1}{2} + iE_\perp/\mathrm{rate}_t$ and $-\frac{1}{2} + iE_\perp/\mathrm{rate}_t$. The recurrence relations of the cylinder functions connect two functions whose orders differ by one, and with them one verifies by substitution that

$$\hat{\psi} = \sqrt{u_d}\left[Z_+(u_d)\, w + i\, Z_-(u_d)\, \alpha^d w\right] \tag{23}$$

solves Eq. (19) for each of the eight w. Two independent kinds of cylinder function give all sixteen solutions. Equation (23) was verified with the Bessel functions of an arbitrary-precision library, to a relative residual of 10^{-15}.

C. The deflation creates particles

At early times, $u_d \to 0$, a Bessel function of the first kind behaves as the power of its argument that its order gives. The solution that has positive energy at early times, $\propto e^{-iE_\perp x4}\, w$, is therefore Eq. (23) with Z_+ the Bessel function of the first kind of the order $-\frac{1}{2} - iE_\perp/\mathrm{rate}_t$ and Z_- minus that of the order $+\frac{1}{2} - iE_\perp/\mathrm{rate}_t$, a pair that obeys the same recurrence relations. At late times, $u_d \to \infty$, the solution of positive energy is Eq. (23) with the Hankel functions of the second kind, which behave as e^{-iu_d}, and the solution of negative energy is the one with the Hankel functions of the first kind. The connection formula of the Bessel functions expresses the first solution through the other two, and the weight of the negative-energy part divided by that of the positive-energy part is $e^{-2\pi E_\perp/\mathrm{rate}_t}$. A mode that starts empty therefore ends with

$$n_{\mathrm{cr}} = \frac{1}{e^{2\pi E_\perp/\mathrm{rate}_t} + 1} \tag{24}$$

particles in each of its eight internal states, and with as many antiparticles. This is the Fermi-Dirac distribution in the energy $E_\perp$ at the temperature

$$T_{\mathrm{defl}} = \frac{\mathrm{rate}_t}{2\pi}. \tag{25}$$

For equal rates it is the temperature that Gibbons and Hawking found for de Sitter space [10]; the creation of particles by a time-dependent metric is the phenomenon discovered by Parker [9, 11].

The numerical solution of Eq. (19) for all sixteen components confirms the law (Fig. 3). For $E_\perp/\mathrm{g55}$ between 0.25 and 3 the created number agrees with Eq. (24) to a relative deviation of at most 3.2×10^{-6}, while the numbers themselves range from 0.17 to 6.5×10^{-9}; the eight internal states are created equally. With three equal rates the result depends neither on the direction of the wavenumber nor on the constants const_t. With unequal rates no exact solution is known. If the second physical wavenumber is still below $e^{-3}E_\perp$ when the first reaches $E_\perp$, the mean created number is Eq. (24) with the rate of the coordinate that becomes energetic first, to within 1.2 per cent. If two directions become energetic together, the eight internal states are created *unequally*, in the cases computed by up to a factor 110 between the most and the least populated state, and the result is known only numerically.

D. What the result means

Each mode with $k_t \neq 0$ is populated once, at its crossing. Its particles then gain energy without bound, as $(E_\perp^2 + k_d^2)^{1/2}$, like the classical particle of Sec. II E. Modes with $k_t = 0$ stay empty. Two remarks are in order.

First, the stress of the created particles must eventually change the metric. That reaction of the fields on the metric has not been computed, and the results of this section hold for as long as it can be neglected. For any rate of deflation that the bounds of Sec. XI allow today the temperature (25) is below 10^{-30} K, and nothing is created.

Second, the instantaneous energy of a quantum in a mode with $k_t \neq 0$ is at least $|k_t|/b_t$, which grows as $e^{\mathrm{rate}_t\, x4}$. A process with a given energy at its disposal can excite such a mode only before the mode has crossed that energy. If the deflating coordinates are periodic, the smallest wavenumber that differs from zero is fixed, and the sector of fields that do not depend on $x5, x6, x7$ is separated from the rest by a gap that grows exponentially. This is the sense in which the deflation supports the restriction to that sector, which the earlier paper assumed. If the deflating coordinates are infinite, modes of arbitrarily small k_t exist at every time, and the restriction remains an assumption.

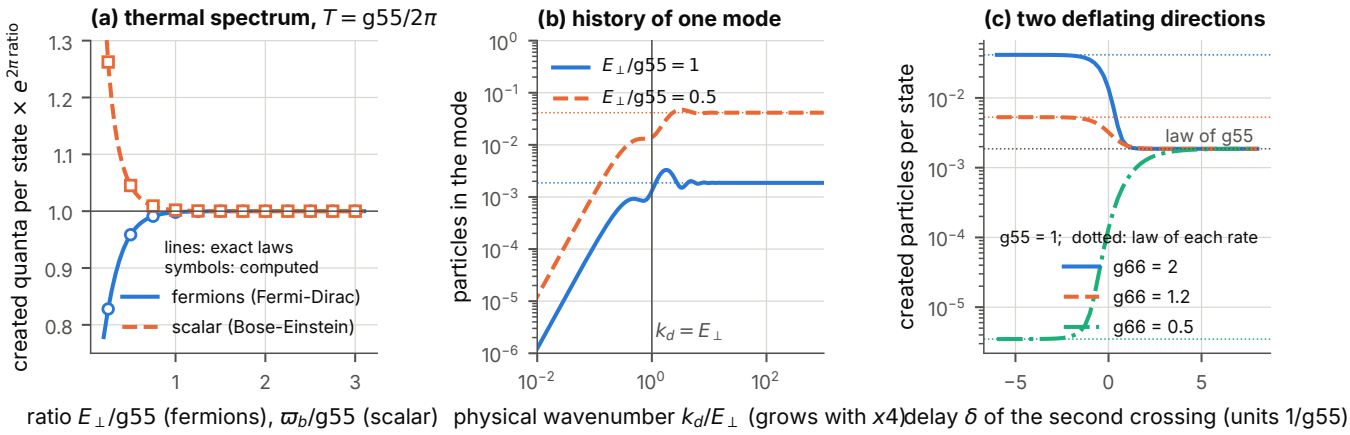


FIG. 3. Creation of particles on the deflating 7+1 spacetime. (a) Created quanta per state, multiplied by $e^{2\pi\,\mathrm{ratio}}$, against the ratio of the energy to the rate. Fermions follow the Fermi-Dirac law (24), the modes of the Vorton-Scalar the Bose-Einstein law (35), both at the temperature $\mathrm{g55}/2\pi$. Lines: the exact laws; symbols: computed. (b) History of two modes, with $E_\perp/\mathrm{g55} = 1$ and 0.5: the number of particles in the mode against the physical wavenumber $k_d/E_\perp$, which grows with the time. After the crossing at $k_d = E_\perp$ the number settles at the value of Eq. (24) (dotted). (c) Two deflating directions, $\mathrm{g55} = 1$ and $\mathrm{g66} = 2$, 1.2 or 0.5: mean number of created particles per state against the delay between the two crossings; the dotted lines are Eq. (24) with each of the two rates.

E. The numerical method

The equations of this section are linear, with coefficients that grow exponentially, and an integrator with a fixed step fails on them. The program uses a commutator-free Magnus method of the fourth order: over each step the solution is advanced by the product of two matrix exponentials of linear combinations of the coefficient matrix at two Gauss points. Because the square of h is a multiple of the unit matrix at any one time, each exponential is evaluated in closed form, and every step is exactly unitary. The length of a step is chosen so that neither the phase accumulated in it nor the relative change of the coefficients exceeds a fixed bound. The number of particles is read off by projection on the eigenstates of the Hamiltonian corrected to the first order in the rate at which its direction changes, which removes the leading spurious oscillation of the particle number at finite time. An independent program, which uses an explicit Runge-Kutta method of the eighth order and the Bessel functions of another library, reproduces Eq. (24) to 2.2×10^{-5} [4].

V. THE VORTON-SCALAR AND THE DEFLATION CLOCK

A. The model

The earlier paper [2] introduced a complex scalar field, the *Vorton-Scalar* Φ_{VS}, whose vacuum value gives the fermions their mass and whose exchange binds two fermions into a pair. We restate the model on the deflating spacetime. It has three real, positive constants: the self-coupling λ_{VS} of the scalar, the constant v_{VS} that fixes the magnitude $v_{\mathrm{VS}}/\sqrt{2}$ at which the potential of the scalar is lowest, and the Yukawa coupling y_{VS} of the scalar to the fermions. The chirality matrix is denoted by χ, with $\chi = i\,\gamma^0\gamma^1\cdots\gamma^7$, and the chiral projectors are $P^{L,R} = \frac{1}{2}(1 \mp \chi)$.

On a curved spacetime a Lagrangian is multiplied by the volume factor V_d, derivatives of spinors become covariant derivatives, and the flat metric is replaced by the curved one. For the Lagrangian of the earlier paper this gives

$$\begin{aligned}\mathcal{L}^{\mathrm{VS}}_{1671d} = V_d \Big[& \bar\psi\, i\gamma^\mu \nabla_\mu \psi + g^{\mu\nu}\,\partial_\mu \Phi^*_{\mathrm{VS}}\,\partial_\nu \Phi_{\mathrm{VS}} \\ & - \lambda_{\mathrm{VS}} \Big(|\Phi_{\mathrm{VS}}|^2 - \frac{v^2_{\mathrm{VS}}}{2} \Big)^2 \\ & - y_{\mathrm{VS}}\, \bar\psi \big(\Phi_{\mathrm{VS}}\, P^R + \Phi^*_{\mathrm{VS}}\, P^L \big) \psi \Big], \end{aligned} \tag{26}$$

with $\bar\psi = \psi^\dagger \beta$. The Lagrangian has two phase symmetries, as on flat spacetime, because their transformations do not involve the coordinates: the fermion number $U(1)_F$, $\psi \to e^{i\epsilon}\psi$, and the chiral symmetry $U(1)_{\mathrm{VS}}$, $\psi \to e^{i\epsilon\chi}\psi$ with $\Phi_{\mathrm{VS}} \to e^{-2i\epsilon}\Phi_{\mathrm{VS}}$.

A constant field $\Phi_{\mathrm{VS}} = v_{\mathrm{VS}}/\sqrt{2}$ with $\psi = 0$ solves the field equations on the deflating spacetime, because a constant has no derivatives: the vacuum value is the same at all times. It breaks $U(1)_{\mathrm{VS}}$. Around it the Vorton-Scalar has two modes: the *radial mode*, a real field of the mass m_s, and the *phase mode*, a real massless field, the Goldstone boson of the broken symmetry. The fermions acquire the mass m. Both modes couple to the fermions with the same strength g_{VS}, the radial mode to the scalar density $\bar\psi\psi$ and the phase mode to the pseudoscalar den-

sity $\bar\psi\, i\chi\, \psi$:

$$m = \frac{y_{\rm VS}\, v_{\rm VS}}{\sqrt{2}}, \qquad m_s = \sqrt{2\lambda_{\rm VS}}\, v_{\rm VS},$$
$$g_{\rm VS} = \frac{m}{v_{\rm VS}}, \qquad \alpha_{\rm VS} = \frac{g_{\rm VS}^2}{4\pi}. \tag{27}$$

B. Fields that do not depend on the extra coordinates

As in the earlier paper, consider fields that do not depend on the four extra coordinates $x0, x5, x6, x7$, and introduce the weighted fields

$$\hat\psi = \sqrt{V_d}\,\psi, \qquad \hat\Phi_{\rm VS} = \sqrt{V_d}\,\Phi_{\rm VS}. \tag{28}$$

Each term of Eq. (26) is rewritten in turn. The kinetic term of the fermions is the flat one for $\hat\psi$ (Sec. III). The Yukawa term contains one scalar and two fermion fields and therefore the factor $V_d \cdot V_d^{-3/2}$. The potential becomes $(\lambda_{\rm VS}/V_d)\,(|\hat\Phi_{\rm VS}|^2 - V_d v_{\rm VS}^2/2)^2$. In the kinetic term of the scalar the derivative along the time acts on the weight as well, $\sqrt{V_d}\,\partial_4\Phi_{\rm VS} = \partial_4\hat\Phi_{\rm VS} + \frac{1}{2}{\rm gsum}\,\hat\Phi_{\rm VS}$, which gives a term $\frac{1}{4}{\rm gsum}^2|\hat\Phi_{\rm VS}|^2$ and a total derivative. The result is the Lagrangian of the earlier paper for the weighted fields, with three replacements and one additional term:

$$y_{\rm VS} \to \frac{y_{\rm VS}}{\sqrt{V_d(x4)}}, \qquad \lambda_{\rm VS} \to \frac{\lambda_{\rm VS}}{V_d(x4)},$$
$$v_{\rm VS} \to v_{\rm VS}\sqrt{V_d(x4)}, \qquad \text{and } + \frac{{\rm gsum}^2}{4}\,|\hat\Phi_{\rm VS}|^2. \tag{29}$$

The three replacements are those by which the earlier paper passed to the four-dimensional description, where the volume of the extra coordinates was a constant. Here that volume is the coordinate volume of a cell times $V_d(x4)$, and it shrinks. The additional term lowers the square of the mass of each mode of the scalar by ${\rm gsum}^2/4$ in the equation of the weighted field; Section V D shows what it does. From here on the couplings $y_{\rm VS}$, $\lambda_{\rm VS}$, $v_{\rm VS}$, $g_{\rm VS}$ and $\alpha_{\rm VS}$ are those of the four-dimensional description at the time considered.

C. The deflation clock

Define the *deflation clock*

$$\tau_d = {\rm gsum}\,(x4 - x4_{\rm ref}), \tag{30}$$

the number of e-foldings by which the volume V_d has shrunk since a reference time $x4_{\rm ref}$, which is chosen in each application. By Eq. (29) every quantity of the four-dimensional description is proportional to $e^{n\tau_d}$ with a definite exponent n (Table I). Three observations organize what follows.

The masses do not change. The fermion mass m and the mass m_s of the radial mode contain the couplings in combinations from which the volume cancels. So do the Fermi momentum and the Fermi energy of matter with a given number of particles per unit of ordinary volume.

The coupling grows. The coupling $g_{\rm VS} = m/v_{\rm VS}$ grows as $e^{\tau_d/2}$, because the four-dimensional vacuum value falls. The strength of the attraction, $\alpha_{\rm VS}$, grows as e^{τ_d}: it doubles every time the volume of the deflating coordinates halves.

The two numbers that decide the state of matter grow at the same rate. These are

$$D_{\rm VS} = \alpha_{\rm VS}\,\frac{m}{m_s}, \qquad C = \frac{g_{\rm VS}^2 m^2}{m_s^2}. \tag{31}$$

The first decides whether two fermions form a bound pair (Sec. VII), the second whether matter keeps its fermion mass (Sec. IX). Their ratio $C/D_{\rm VS} = 4\pi\, m/m_s$ is fixed.

A model of the earlier paper is specified by the masses and one coupling. On a deflating spacetime it is specified by the masses and the value of the coupling *at one time*; the deflation then carries it through all larger couplings. The constants of the earlier paper become functions of the time, and its thresholds are passed one after another (Fig. 4).

D. Modes of the Vorton-Scalar

Let $\varphi_{\rm bos}$ stand for the radial mode or the phase mode, with the mass $m_{\rm bos} = m_s$ or $m_{\rm bos} = 0$. To first order in the field, a mode with a wavenumber of the magnitude k along the flat coordinates and the wavenumbers k_t along the deflating ones obeys

$$\frac{1}{V_d}\,\partial_4\big(V_d\,\partial_4\varphi_{\rm bos}\big) + \Big(k^2 + \sum_t \frac{k_t^2}{b_t^2} + m_{\rm bos}^2\Big)\varphi_{\rm bos} = 0. \tag{32}$$

Modes that do not depend on the deflating coordinates. With $k_t = 0$ the equation has constant coefficients, and its solutions are the exponentials

$$\varphi_{\rm bos} \propto \exp\Big\{\Big[\frac{\rm gsum}{2} \pm \sqrt{\frac{{\rm gsum}^2}{4} - k^2 - m_{\rm bos}^2}\Big]x4\Big\}. \tag{33}$$

If $k^2 + m_{\rm bos}^2 > {\rm gsum}^2/4$, the mode oscillates with the frequency $\varpi_b = (k^2 + m_{\rm bos}^2 - {\rm gsum}^2/4)^{1/2}$, and its amplitude grows as $e^{{\rm gsum}\,x4/2}$. The growth is that of the weight alone: the weighted field has a constant amplitude, and the number of quanta in a cell of fixed coordinates is conserved while the cell shrinks. Otherwise the mode does not oscillate. For the radial mode this never happens if $m_s > {\rm gsum}/2$, which holds by many orders of magnitude for any rate that observation allows (Sec. XI). For the phase mode it happens at wavelengths longer than $4\pi/{\rm gsum}$. At $k = 0$ the two exponents of

TABLE I. How the quantities of the four-dimensional description follow the deflation clock: each is proportional to $e^{n\tau_d}$. The last two rows hold under additional assumptions, which Secs. XI and XII state and examine.

quantity	meaning	exponent n
V_d	$b_5 b_6 b_7$: the factor $\sqrt{\lvert\det g\rvert}$ of the measure	-1
$y_{\rm VS}$ of the four-dimensional description	Yukawa coupling of the weighted fields	$\frac{1}{2}$
$\lambda_{\rm VS}$ of the four-dimensional description	quartic coupling of the weighted Vorton-Scalar	1
$v_{\rm VS}$ of the four-dimensional description	vacuum value of the weighted Vorton-Scalar	$-\frac{1}{2}$
m	fermion mass $y_{\rm VS} v_{\rm VS}/\sqrt{2}$	0
m_s	mass of the radial mode $\sqrt{2\lambda_{\rm VS}}\, v_{\rm VS}$	0
$g_{\rm VS}$	coupling of the radial mode to the fermions, $m/v_{\rm VS}$	$\frac{1}{2}$
$\alpha_{\rm VS}$	$g_{\rm VS}^2/4\pi$	1
$G_{\rm VS}$	constant of the Fermi type of the neutral-current interaction, $1/(\sqrt{2}\, v_{\rm VS}^2)$	1
$D_{\rm VS}$	$\alpha_{\rm VS}\, m/m_s$: decides whether two fermions bind	1
C	$g_{\rm VS}^2 m^2/m_s^2$: decides the state of matter	1
vortex energy per length	proportional to $v_{\rm VS}^2$ (four-dimensional)	-1
$G_{\rm Fermi}$ (Fermi constant of the weak interaction) if the vacuum value of the Vorton-Scalar sets the electroweak scale (assumption HV)	$1/(\sqrt{2}\, v_{\rm VS}^2)$, four-dimensional vacuum value	1
G_N (Newton's constant) if gravity is eight-dimensional (A2)	G_8 divided by the volume of $x5, x6, x7$	1

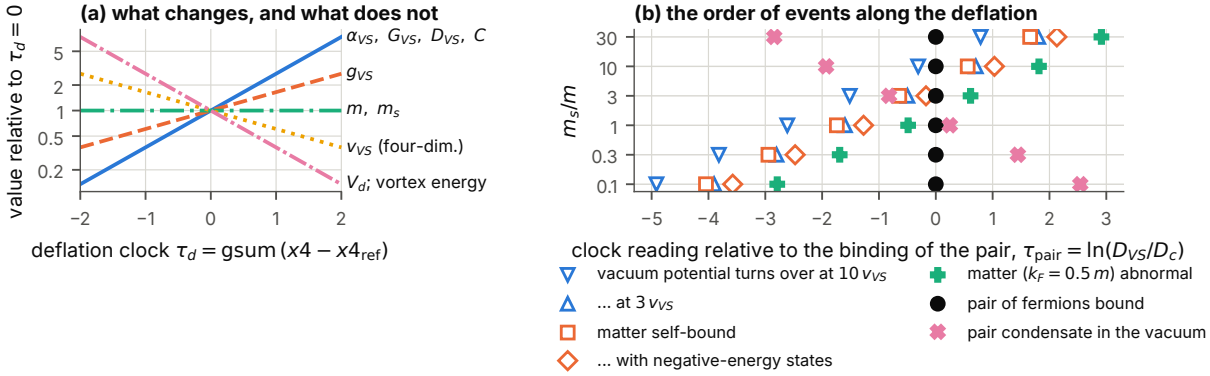


FIG. 4. The deflation clock. (a) How the quantities of the four-dimensional description change with the clock τ_d (logarithmic scale; values relative to $\tau_d = 0$). (b) The order of events along the deflation for six ratios m_s/m: the clock reading of each event relative to the moment at which two fermions form a bound pair (Table IV).

the phase mode are 0 and gsum: a constant phase, which costs nothing, and a phase that advances at a rate proportional to $1/V_d$, which is the conservation of the charge of $U(1)_{\rm VS}$ in a shrinking volume. Neither is an instability of the vacuum. Figure 5 shows the exponents.

The potential between two fermions. Two fermions at rest attract each other by exchanging the radial mode. The density of a fixed number of fermions grows as $1/V_d$, so that the radial mode obeys Eq. (32) with a source proportional to $e^{\text{gsum}\, x4}$. The solution that follows the source is $e^{\text{gsum}\, x4}/(k^2 + m_{\rm bos}^2)$ times the source, as substitution shows: the terms $\text{gsum}^2 - \text{gsum}\cdot\text{gsum}$ cancel, and the denominator contains the *unshifted* mass. The potential between two fermions at rest is therefore, at every moment, the Yukawa potential of the earlier paper,

$$V_s(r) = -\alpha_{\rm VS}(x4)\,\frac{e^{-m_s r}}{r}, \tag{34}$$

with the range $1/m_s$ unchanged and the strength of the moment. The numerical solution from rest approaches this form with a transient that decays as $e^{-\tau_d/2}$ [Fig. 5(c)]; after $\tau_d = 20$ the relative difference is below 4×10^{-5} in the four cases computed.

Modes that depend on a deflating coordinate. For $k_t \neq 0$ the weighted field obeys $\partial_4^2\hat{\varphi}_{\rm bos} + (\varpi_b^2 + k_d^2)\,\hat{\varphi}_{\rm bos} = 0$ with the physical wavenumber k_d of Eq. (20), and the scalar behaves as the fermion field does. The frequency grows without bound. When one rate governs, the solutions are Bessel functions of the imaginary order $i\varpi_b/\text{rate}_t$ and of the argument u_d, and a mode that starts empty ends with

$$n_{\rm cr} = \frac{1}{e^{2\pi\varpi_b/\text{rate}_t} - 1} \tag{35}$$

quanta: the Bose-Einstein distribution at the same temperature $T_{\rm defl}$ as for the fermions. The program confirms

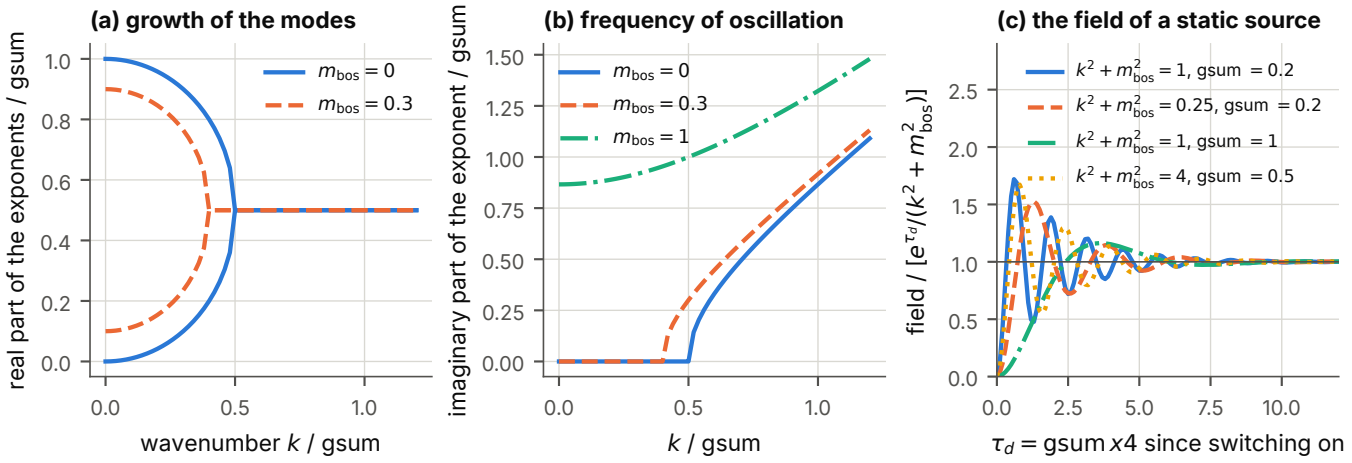


FIG. 5. Modes of the Vorton-Scalar that do not depend on the deflating coordinates. (a) Real parts of the two exponents of Eq. (33) against the wavenumber, for the phase mode and for a radial mode, in units of gsum. (b) Their imaginary part, the frequency of oscillation. (c) The field that a static source builds up from zero, divided by the form $e^{\tau_d}/(k^2+m_{\rm bos}^2)$ that leads to Eq. (34).

Eq. (35) in 36 cases to a relative deviation of 1.9×10^{-5} (Fig. 3).

VI. PAIRING AND THE GAP ALONG THE CLOCK

A. The pair channels

A pair of fermions is created by an operator $\psi^T X\psi$ with a 16×16 matrix X; the choice of X is the *channel* of the pair. The earlier paper classified the channels and computed, for each, the *exchange amplitudes* F_S and F_P of the radial mode and of the phase mode: the matrix elements of the exchange of one boson between two fermions of opposite momenta in the internal state of the channel. It found that the radial mode attracts in every channel, and that the phase mode attracts in the four *vector pairs along the extra axes*,

$$\psi^T C^-\gamma^a\psi, \qquad a\in\{0,5,6,7\}, \tag{36}$$

and repels in their partners that are built with two Dirac matrices. Here $C^-=\gamma^4\gamma^0\gamma^1\cdots\gamma^7$ is the matrix that turns each Dirac matrix into minus its transpose, and the index a labels the four extra axes. The pair along the hidden axis, $a=0$, has spin 0 in ordinary spacetime; it is the *hidden-axis singlet* of hypothesis H1.

The amplitudes are computed from the Dirac matrices of the frame, which the deflation does not change, and each is proportional to the square of the coupling. For fields that do not depend on $x5,x6,x7$ the amplitude of a channel at the time $x4$ is therefore

$$F_{S,P}(x4) = g_{\rm VS}(x4)^2 \times \big(F_{S,P}\text{ of the flat spacetime at } g_{\rm VS}=1\big). \tag{37}$$

The program computes the amplitudes of the 15 representative channels of the earlier paper again from the matrices of the deflating spacetime, at three momenta and 13 scattering angles; the numbers agree with those of the earlier paper to every digit recorded there, and the amplitudes of the hidden-axis channel reproduce its closed forms to 1.1×10^{-15}. The conclusion of the earlier paper about the channels therefore stands at every moment (Fig. 6): the most attractive channel is the vector pair along an extra axis.

B. The gap grows much faster than the coupling

In matter of a given density the attraction opens a gap Δ_F at the Fermi momentum k_F: the energy that it costs to break a pair. The gap equation of the earlier paper, which generalizes that of Bardeen, Cooper and Schrieffer [12], holds at each moment with the couplings of that moment. Its solution as a function of the clock is the *adiabatic history* of the paired state: the state that matter would have if it were always in equilibrium. (Section VIII examines when it can be.) The strength of the pairing is measured by the dimensionless number $\lambda = N(0)\,W(k_F,k_F)$, the density of states at the Fermi surface times the kernel of the gap equation there; in the vector channel it is the sum $\lambda_S+\lambda_P$ of the contributions of the two modes. Since the kernel is proportional to $g_{\rm VS}^2$,

$$\lambda \;\propto\; e^{\tau_d}. \tag{38}$$

The program solves the gap equation at 34 moments for each of four sets of the mass m_s and the Fermi momentum (Table II, Fig. 7). There are three results.

The order of the channels does not change. At every moment the gap of the vector channel is at least as large as the gap that the radial mode alone would give, and

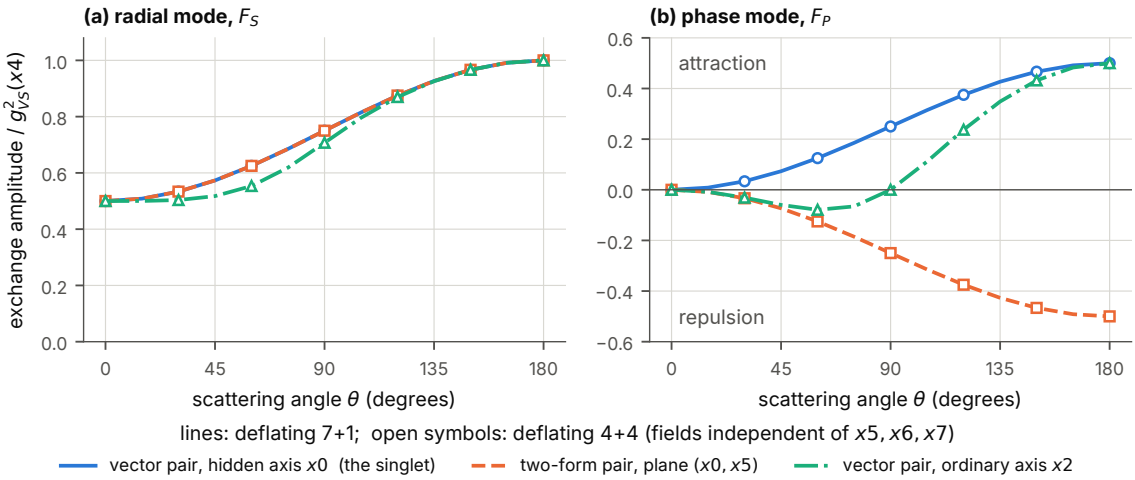


FIG. 6. Exchange amplitudes in three pair channels, divided by the square of the coupling of the moment, against the scattering angle, for momenta of the magnitude m in the plane of $x1$ and $x2$. Lines: deflating 7+1 spacetime; open symbols: deflating 4+4 spacetime. (a) The radial mode attracts in all three channels. (b) The phase mode attracts in the vector pair along the hidden axis and repels in its partner built with two Dirac matrices. The amplitudes are those of flat spacetime: the deflation changes their overall strength and nothing else.

TABLE II. The gap along the deflation clock for two of the four sets computed. $g_{\rm VS}$ and λ: coupling and pairing strength of the moment. E_F: Fermi kinetic energy. T_c: critical temperature. Rate of growth: $d\ln\Delta_F/d\tau_d$; the last column, its product with λ, is given where the gap is between 10^{-6} and 0.02 of E_F.

m_s/m, k_F/m	τ_d	$g_{\rm VS}$	λ	Δ_F/E_F	T_c/Δ_F	rate of growth	$\lambda\times$ rate
0.10, 0.50	−0.63	0.730	0.062	1.17×10^{-7}	0.567	—	—
0.10, 0.50	−0.13	0.939	0.102	7.40×10^{-5}	0.567	9.88	1.01
0.10, 0.50	0.38	1.208	0.169	3.70×10^{-3}	0.567	6.01	1.01
0.10, 0.50	0.88	1.554	0.279	0.0407	0.565	3.74	—
0.10, 0.50	1.39	2.000	0.463	0.189	0.555	2.50	—
0.10, 0.50	1.89	2.573	0.766	0.563	0.528	1.96	—
0.10, 0.50	2.39	3.311	1.268	1.49	0.480	1.96	—
1.0, 0.50	0.88	1.554	0.056	1.02×10^{-7}	0.567	—	—
1.0, 0.50	1.13	1.763	0.072	5.62×10^{-6}	0.567	14.02	1.01
1.0, 0.50	1.39	2.000	0.092	1.27×10^{-4}	0.567	10.93	1.01
1.0, 0.50	1.64	2.269	0.119	1.45×10^{-3}	0.567	8.54	1.02
1.0, 0.50	1.89	2.573	0.153	9.76×10^{-3}	0.567	6.71	1.03
1.0, 0.50	2.14	2.919	0.197	0.0439	0.566	5.33	—
1.0, 0.50	2.39	3.311	0.253	0.147	0.563	4.37	—

that at least as large as the gap of the channel in which the phase mode repels. The gap never decreases.

The critical temperature follows the gap. Wherever the gap is below 5 per cent of the Fermi kinetic energy E_F, the critical temperature is $T_c = 0.567\,\Delta_F$, the ratio of weak coupling, to within 0.002.

The law of growth. In weak coupling the gap is proportional to $e^{-1/\lambda}$. With $\lambda \propto e^{\tau_d}$,

$$\frac{d\ln\Delta_F}{d\tau_d} = \frac{1}{\lambda}. \tag{39}$$

In the computed histories the product $\lambda\, d\ln\Delta_F/d\tau_d$ lies between 1.00 and 1.03 wherever the gap is between 10^{-6} and 0.02 of E_F.

Equation (39) is the central quantitative statement about the superfluid on a deflating spacetime. While the coupling grows by one e-folding, the gap of a weakly paired state with $\lambda = 0.1$ grows by ten e-foldings, a factor of twenty thousand. A paired state that is unobservably weak at one time is strong a few e-foldings of the volume later.

VII. THE BOUND PAIR AND THE ORDER OF EVENTS

A. The moment of binding

Two fermions in the Yukawa potential (34) form a bound state if and only if the number $D_{\rm VS}$ of Eq. (31)

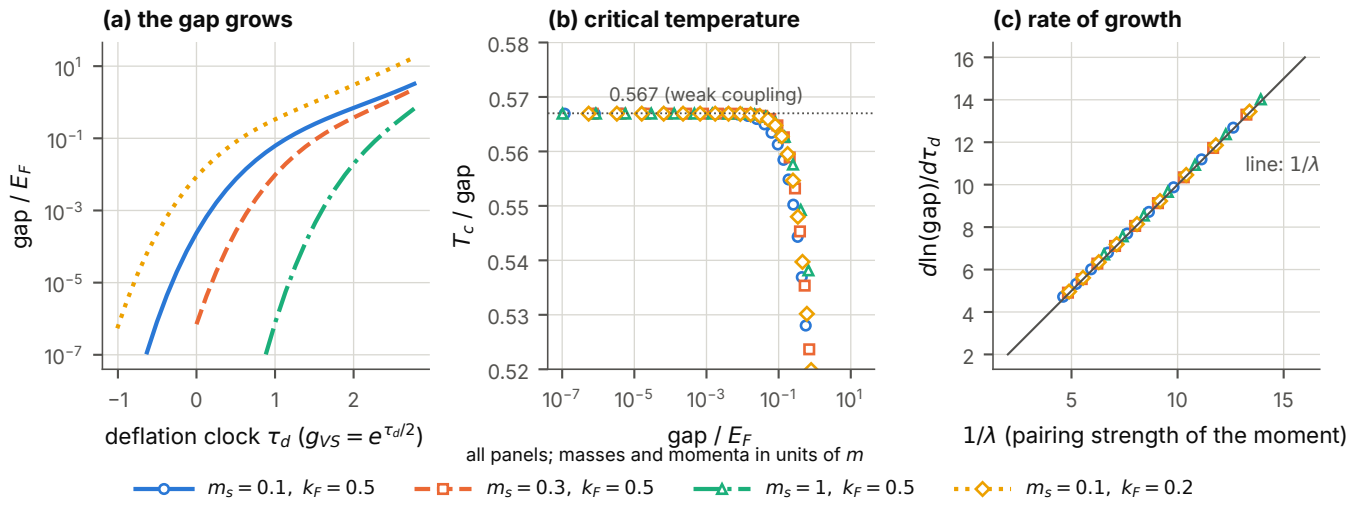


FIG. 7. The gap along the deflation clock, for four sets of the mass of the radial mode and of the Fermi momentum (in units of m). (a) The gap of the vector channel in units of the Fermi kinetic energy. (b) The ratio of the critical temperature to the gap. (c) The rate of growth of the gap against $1/\lambda$; the line is Eq. (39).

TABLE III. Two fermions as the clock passes the moment of binding: the scattering length in units of $1/m_s$, the number of bound states, and the binding energy in units of m_s^2/m.

$\tau_{\rm pair}$	$D_{\rm VS}$	$a_{\rm sc}\, m_s$	bound states	$E_b\, m/m_s^2$
-0.105	1.5118	-12.50	0	0
-0.095	1.5279	-13.93	0	0
-0.010	1.6630	-134.0	0	0
0.010	1.6966	135.9	1	5.50×10^{-5}
0.486	2.7314	2.698	1	0.232
0.994	4.5409	-0.1288	1	1.82

exceeds a critical value, $D_c = 1.67981$. On a deflating spacetime $D_{\rm VS} \propto e^{\tau_d}$, so that the pair becomes bound at a definite time. Measure the clock from that moment:

$$\tau_{\rm pair} = \ln \frac{D_{\rm VS}}{D_c} = \text{gsum}\left(x4 - x4_{\rm binding}\right). \qquad (40)$$

Table III and Fig. 8 show the scattering length $a_{\rm sc}$ of two fermions and the binding energy E_b of the pair as the clock passes the binding.

The scattering length. Just before the binding the scattering length is large and negative, just after it large and positive: $-134.0/m_s$ at $0.99\, D_c$ and $+135.9/m_s$ at $1.01\, D_c$. At the moment of binding it is infinite: the two fermions are at a scattering resonance. *Every pair of fermions of the model passes through this resonance once, at the same time.* Section XI examines what that means if the fermions are the dark matter.

The binding energy. Just after the binding the binding energy starts quadratically,

$$E_b \;\propto\; \left(e^{\tau_{\rm pair}} - 1\right)^2 \frac{m_s^2}{m}, \qquad (41)$$

with the fitted exponent 2.01 between 2 and 8 per cent above D_c. Because E_b vanishes at the threshold, the time $1/E_b$ that the pair needs to adjust is infinite there. However slow the deflation is, the pair cannot follow it adiabatically at first.

The same threshold from the gap equation. In vacuum the linearized gap equation is the equation of a bound state of two fermions, with a relativistic kernel. It reaches the eigenvalue 1 at $D_{\rm VS} = 1.7347$, 1.6902 and 1.6810 for $m_s/m = 0.3$, 0.1 and 0.03, approaching the value D_c of the Schrödinger equation as the radial mode becomes light and the binding non-relativistic.

B. The order of events

The other thresholds of the earlier paper are thresholds in the number C of Eq. (31), which grows at the same rate as $D_{\rm VS}$. Because $C/D_{\rm VS} = 4\pi m/m_s$ is fixed, the order in which the events happen depends on the ratio m_s/m alone. Table IV gives the clock reading $\tau_{\rm pair}$ of each event; a negative number means that the event precedes the binding of the pair. The events other than the binding are derived in Sec. IX.

For a *light radial mode*, $m_s/m \le 3$ in the table, the pair becomes bound only *after* the one-loop potential of the Vorton-Scalar has fallen below its vacuum value at a field of three times the vacuum value: the deflation endangers the vacuum of the model before it binds the pair. The condition for the opposite order is $4\pi D_c\, m/m_s < 4.27$, that is, $m_s/m > 4.9$. For a *heavy radial mode*, $m_s/m = 10$ or 30, the pair becomes bound first, and matter becomes self-bound and loses its fermion mass later. Nothing in the model decides between the two orders; the table states what follows from each.

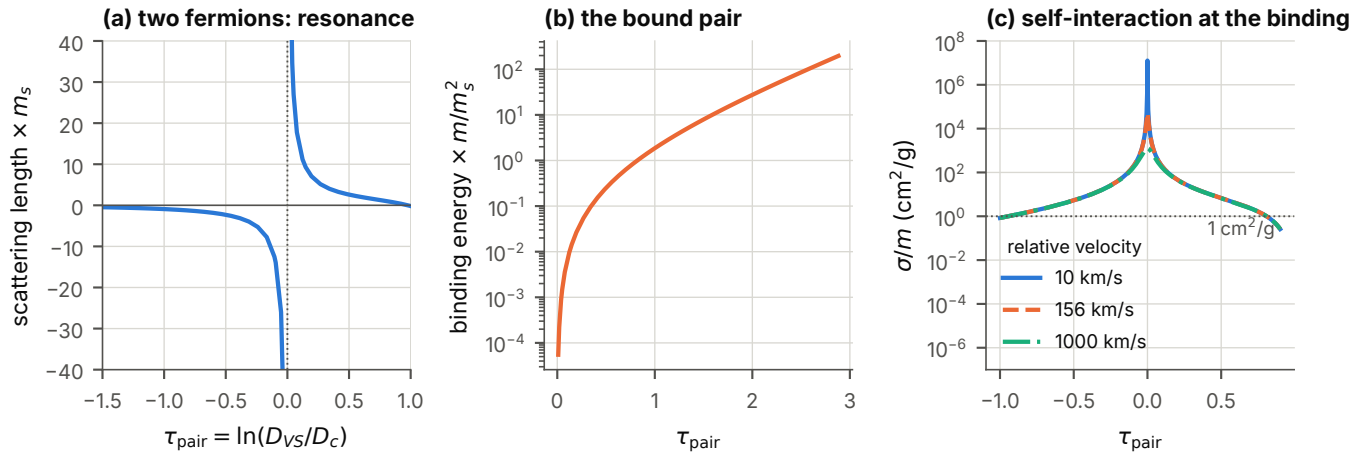


FIG. 8. The bound pair along the clock. (a) Scattering length of two fermions in units of $1/m_s$ against $\tau_{\rm pair}$: the resonance at the moment of binding. (b) Binding energy of the pair. (c) Cross-section per mass of two fermions at three relative velocities while the clock passes the binding, for the example particle of Sec. XI F.

TABLE IV. The order of events along the deflation: the clock reading $\tau_{\rm pair}$ of each event, relative to the moment at which two fermions form a bound pair, for six ratios of the mass m_s of the radial mode to the fermion mass m.

event	threshold	0.1	0.3	1	3	10	30
the one-loop potential of the Vorton-Scalar falls below its vacuum value at the field $10\,v_{\rm VS}$	$C = 1.5477$	−4.92	−3.82	−2.61	−1.51	−0.31	0.79
matter is self-bound (Fermi sea only)	$C = 3.7011$	−4.04	−2.94	−1.74	−0.64	0.56	1.66
the one-loop potential falls below its vacuum value at the field $3\,v_{\rm VS}$	$C = 4.2694$	−3.90	−2.80	−1.60	−0.50	0.70	1.80
matter is self-bound (with the negative-energy states)	$C = 5.9218$	−3.57	−2.48	−1.27	−0.17	1.03	2.13
matter of Fermi momentum $0.5\,m$ becomes abnormal (massless fermions)	$C = 12.9713$	−2.79	−1.69	−0.49	0.61	1.82	2.91
two fermions form a bound pair	$D_{\rm VS} = 1.6798$	0.00	0.00	0.00	0.00	0.00	0.00
pair condensate in the vacuum (ladder equation, cut-off $100\,m$)	$\alpha_{\rm VS} = 2.13$ to 2.93	2.54	1.44	0.25	−0.82	−1.93	−2.84

VIII. CAN THE PAIRED STATE FOLLOW THE DEFLATION?

A. The question, and the parameter that decides it

Section VI gave the gap that the paired state would have in equilibrium at each moment. But a state can change at most at the rate of its own energy scale, which for a paired state is the gap. By Eq. (39) the equilibrium gap, written $\Delta_{\rm eq}$ in this section, changes at the rate gsum/λ. The paired state can follow only if that rate is small compared with the gap itself:

$$\Xi_d = \frac{\text{gsum}}{\lambda\,\Delta_{\rm eq}} \ll 1. \tag{42}$$

Early in the deflation the gap is exponentially small, and the condition fails however small the rate is. A system that is carried through a change faster than it can respond falls out of equilibrium, as in the formation of defects at a phase transition [13, 14]. The question is what the paired state does then, and when it catches up.

B. The model

The question is answered with the standard reduced model of a superfluid, whose dynamics without collisions is known in detail [15]. Pairs of states of opposite momenta, with the single-particle energy ξ_k measured from the Fermi energy, fill a shell $-E_{\rm sh} < \xi_k < E_{\rm sh}$ with a constant density of states, and all attract one another with the dimensionless strength λ. In the mean-field approximation each pair of states is described by its pair correlation κ, a complex number, and its occupation n_k; the order parameter is $\Delta = \lambda \int \kappa\, d\xi_k$. The equations of motion are

$$\begin{aligned} \frac{d\kappa}{dx4} &= -i\left[2\,\xi_k\,\kappa - \Delta\,(1-2n_k)\right], \\ \frac{dn_k}{dx4} &= 2\,\mathrm{Im}\big(\Delta^*\kappa\big). \end{aligned} \tag{43}$$

They conserve $|2\kappa|^2 + (1-2n_k)^2 = 1$ for every pair of states. In equilibrium $\kappa = \Delta/2E_k$ and $n_k = \frac{1}{2}(1-\xi_k/E_k)$ with the quasi-particle energy $E_k = (\xi_k^2+\Delta^2)^{1/2}$, and the

gap equation gives

$$\Delta_{\rm eq}(\lambda) = \frac{E_{\rm sh}}{\sinh(1/\lambda)} \;\approx\; 2\,E_{\rm sh}\,e^{-1/\lambda} \quad (\lambda \ll 1). \qquad (44)$$

The model keeps what matters for the question, a gap that depends exponentially on a coupling that grows exponentially, and omits the dependence of the gap on the momentum; $E_{\rm sh}$ stands for the range of energies over which the attraction acts. The program starts the model in equilibrium at $\lambda = 0.15$, lets the coupling grow as $\lambda = 0.15\,e^{\tau_d}$ up to $\lambda = 0.5$, and integrates Eqs. (43) for 2000 energies. Doubling the number of energies and halving the step changes no digit of the results quoted. Besides $|\Delta|$ the program records the density $n_{\rm ex}$ of *broken pairs*: the number of pairs of states, per unit of $E_{\rm sh}$ times the density of states, that are not in the ground state that belongs to the order parameter of the moment. In equilibrium it is zero.

C. Results

Table V and Fig. 9 show three results.

The faster the deflation, the further the order parameter falls behind. For a rate of $10^{-4}\,E_{\rm sh}$ the order parameter drops to 0.958 of the equilibrium gap of the moment before it recovers; for $10^{-1}\,E_{\rm sh}$ it drops to 0.098.

The state catches up when Ξ_d has fallen to a value between 0.23 and 0.52. In six of the seven runs the order parameter returns to within 10 per cent of the equilibrium gap at such a value. In the fastest run it has not caught up when the coupling reaches 0.5. This confirms the estimate (42) and fixes the number that "small" means.

The lag leaves broken pairs behind. Their density at the end grows with the rate, from 6.0×10^{-6} to 4.1×10^{-2} in the six slower runs. They are excitations of the superfluid that the deflation itself has made; in a system without dissipation they remain.

Because $\Delta_{\rm eq}$ grows as $e^{-1/\lambda}$ while λ grows only as e^{τ_d}, the parameter Ξ_d falls steeply with time. For any rate there is a moment at which Ξ_d passes through the window of the second result. Before it the paired state is out of equilibrium, with an order parameter below the equilibrium value; after it the adiabatic history of Sec. VI applies. The slower the deflation, the earlier that moment and the smaller the gap at which it happens: it is the moment at which the equilibrium gap has grown to between two and four times gsum/λ.

IX. THE EFFECTIVE ACTION ALONG THE CLOCK

A. The method, and why it carries over

The earlier paper derived its results on the paired state, on matter, on temperature and on the vacuum from the effective action for composite operators of Cornwall, Jackiw and Tomboulis (CJT) [16]: a functional of the expectation value ϕ of the Vorton-Scalar and of the full propagators, whose two-particle-irreducible part is truncated at two loops. Its stationarity conditions are the equation for the fermion mass in matter and the gap equation. On the deflating spacetime, and for fields that do not depend on the extra coordinates, nothing in this construction changes, for two reasons established above. The weighted fermion field obeys the flat equations, so that its propagator is the flat one. And the interaction between the fermions is, at each moment, the exchange of the two modes of the scalar with the coupling of that moment and with the unshifted range (Sec. V D).

The one new term, the shift $-\text{gsum}^2/4$ of the squares of the boson masses in Eq. (29), is smaller than every other scale of the calculation by the square of the ratio of the rate of deflation to a particle mass. For the largest rate that observation allows (Sec. XI) and a mass of 1 keV that ratio is below 10^{-36}. The term is neglected in this section; this is the only approximation made here beyond those of the earlier paper. The test of the truncation on a solvable model in zero dimensions, which does not involve the spacetime, was executed again: the two-loop truncation has a solution for every coupling from 0.01 to 30 and stays within 10.7 per cent of the exact propagator.

Every result of this section is a statement about a *sequence of equilibrium states*. Whether a system can follow the sequence was examined, for the paired state, in Sec. VIII.

B. The paired state

The CJT potential $V[\Delta]$ of the paired state is the effective action per unit volume, measured from the unpaired state, as a functional of the gap function $\Delta(k)$. The program evaluates it at 13 moments, for two choices of the mass m_s at the Fermi momentum $k_F = 0.5\,m$, in the vector channel (Table VI, Fig. 10).

The solution of the gap equation is a minimum at every moment. Newton's method applied to the potential converges to the gap function that the iteration of the gap equation finds, to 7.7×10^{-12}, and the smallest eigenvalue of the second variation is positive throughout. The temperature at which the second variation at $\Delta = 0$ loses its positivity equals the temperature at which the linearized gap equation has the eigenvalue 1, to 2.6×10^{-12}: two routes to the critical temperature agree.

The condensation energy deepens. It is within 15 per cent of the weak-coupling value wherever the gap is below $0.1\,E_F$, and in weak coupling its logarithmic slope along the clock is twice that of the gap, by Eq. (39):

$$\frac{d\ln|V|}{d\tau_d} = \frac{2}{\lambda}. \qquad (45)$$

The pairs shrink. The coherence length of the first set falls from 25 to 1.0 in units of $1/k_F$. At its last five

TABLE V. The paired state while the pairing strength grows from 0.15 to 0.5, for seven rates of deflation: the parameter Ξ_d at the start; the smallest ratio of the order parameter to the equilibrium gap of the moment; the clock reading at which the order parameter has returned to within 10 per cent of the equilibrium gap, and Ξ_d at that moment; the ratio at the end; and the density of broken pairs at the end.

gsum/E_{sh}	Ξ_d at the start	smallest $\lvert\Delta\rvert/\Delta_{\mathrm{eq}}$	τ_d of the catch-up	Ξ_d there	final $\lvert\Delta\rvert/\Delta_{\mathrm{eq}}$	n_{ex} at the end
1×10^{-4}	0.262	0.958	0.020	0.23	1.0000	6.0×10^{-6}
3×10^{-4}	0.786	0.890	0.069	0.47	0.9998	5.1×10^{-5}
1×10^{-3}	2.62	0.737	0.234	0.52	0.9980	4.4×10^{-4}
3×10^{-3}	7.86	0.539	0.454	0.44	0.9886	2.4×10^{-3}
0.01	26.2	0.328	0.780	0.32	0.9347	0.012
0.03	78.6	0.189	1.172	0.24	0.8076	0.041
0.1	262	0.098	—	—	0.1189	0.022

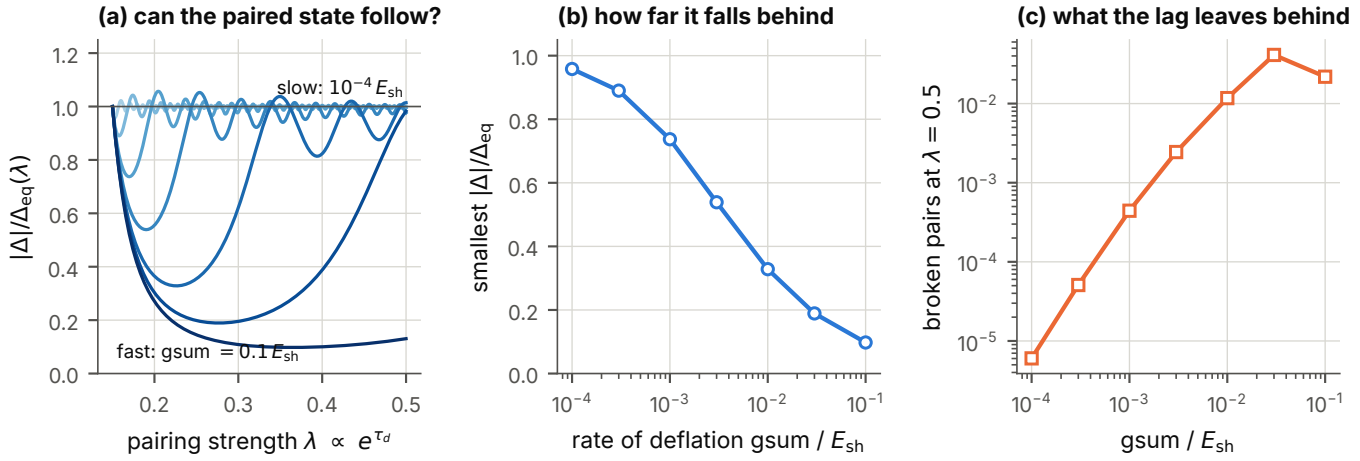


FIG. 9. The paired state while the coupling grows. (a) The order parameter divided by the equilibrium gap of the moment, against the pairing strength, for seven rates of deflation. (b) The smallest value of that ratio against the rate. (c) The density of broken pairs at the end of the run against the rate.

moments the gap exceeds $0.1\,E_F$, and at the last the pair is as small as the distance between the particles. *The deflation carries the paired state from overlapping pairs toward bound pairs.* On flat spacetime this crossover [17, 18] is reached by choosing a larger coupling; here it is reached by waiting.

The universal ratios hold while the coupling is weak. While the critical temperature rises by a factor of sixteen, from 6.3×10^{-3} to 0.10 of the Fermi energy, the ratio of the gap at zero temperature to the critical temperature stays within 0.019 of the weak-coupling value 1.7639, and the relative jump of the heat capacity at T_c within 0.048 of 1.4261. The coefficients of the Ginzburg-Landau energy agree with their weak-coupling formulas to 1.0 per cent while T_c is below $0.05\,E_F$ and differ by up to 7.9 per cent at $T_c = 0.10\,E_F$. The coherence length of the Ginzburg-Landau energy, $0.1333\,v_F/T_c$ in weak coupling with the Fermi velocity v_F, shrinks from 80 to 5.1 in units of $1/m$: the cores of the vortices of the superfluid shrink as the deflation proceeds.

C. Matter

Matter of fixed density loses its fermion mass. In matter the expectation value of the Vorton-Scalar is reduced by the scalar density of the fermions, and with it the fermion mass in the matter, m^*. The earlier paper showed that the reduction depends on the density and on the single number C, and that above a critical value of C the mass vanishes: the matter is then in the *abnormal* state of Lee and Wick [19], a gas of massless fermions in a region without expectation value of the scalar. On the deflating spacetime $C \propto e^{\tau_d}$ grows at fixed density. The critical values are $C = 151.892$, 12.971, 2.556 and 0.617 for the Fermi momenta $k_F = 0.2$, 0.5, 1 and 2 in units of m; they exceed the estimate $\pi^2 m^2/4k_F^2$ of the earlier paper by the factors 2.46, 1.31, 1.04 and 1.00. *Every piece of matter becomes abnormal at a definite moment, the denser the earlier.*

Self-bound matter. Matter is self-bound, that is, its energy per particle at some density lies below the mass of a free fermion, if $C > 3\pi^2/8 = 3.7011$ without the filled negative-energy states and if $C > 3\pi^2/5 = 5.9218$ with them. Once the threshold is passed, the lowest energy per

TABLE VI. The paired state along the clock from the CJT potential, for $k_F = 0.5\,m$. λ: pairing strength. V: the potential at the solution, the condensation energy. Ratio: V divided by its weak-coupling value $-\nu N(0)\Delta_F^2/4$, with $\nu = 8$ internal states and the density of states $N(0)$. Curvature: the smallest eigenvalue of the matrix of second derivatives of the potential. Size: the coherence length of the pair in units of $1/k_F$.

m_s/m, k_F/m	τ_d	g_{VS}	λ	Δ_F/m	V/m^4	ratio	curvature	T_c/Δ_F	size $\times k_F$
0.30, 0.50	0.00	1.80	0.216	2.81×10^{-3}	-4.46×10^{-7}	1.000	0.21	0.567	25.4
0.30, 0.50	0.21	2.00	0.266	6.89×10^{-3}	-2.68×10^{-6}	0.998	0.26	0.566	10.3
0.30, 0.50	0.40	2.20	0.322	0.0135	-1.03×10^{-5}	0.992	0.30	0.564	5.3
0.30, 0.50	0.58	2.40	0.383	0.0230	-2.93×10^{-5}	0.981	0.34	0.559	3.1
0.30, 0.50	0.74	2.60	0.450	0.0354	-6.81×10^{-5}	0.961	0.37	0.552	2.0
0.30, 0.50	0.88	2.80	0.522	0.0511	-1.38×10^{-4}	0.932	0.38	0.542	1.4
0.30, 0.50	1.02	3.00	0.599	0.0707	-2.53×10^{-4}	0.895	0.38	0.530	1.0
1.0, 0.50	0.00	2.40	0.133	4.25×10^{-4}	-1.02×10^{-8}	1.000	0.13	0.567	167.6
1.0, 0.50	0.16	2.60	0.156	1.32×10^{-3}	-9.88×10^{-8}	1.000	0.15	0.567	53.9
1.0, 0.50	0.31	2.80	0.181	3.28×10^{-3}	-6.08×10^{-7}	0.999	0.17	0.567	21.7
1.0, 0.50	0.45	3.00	0.208	6.89×10^{-3}	-2.68×10^{-6}	0.998	0.19	0.566	10.3
1.0, 0.50	0.58	3.20	0.237	0.0128	-9.25×10^{-6}	0.994	0.21	0.564	5.6
1.0, 0.50	0.70	3.40	0.267	0.0218	-2.65×10^{-5}	0.987	0.23	0.561	3.3

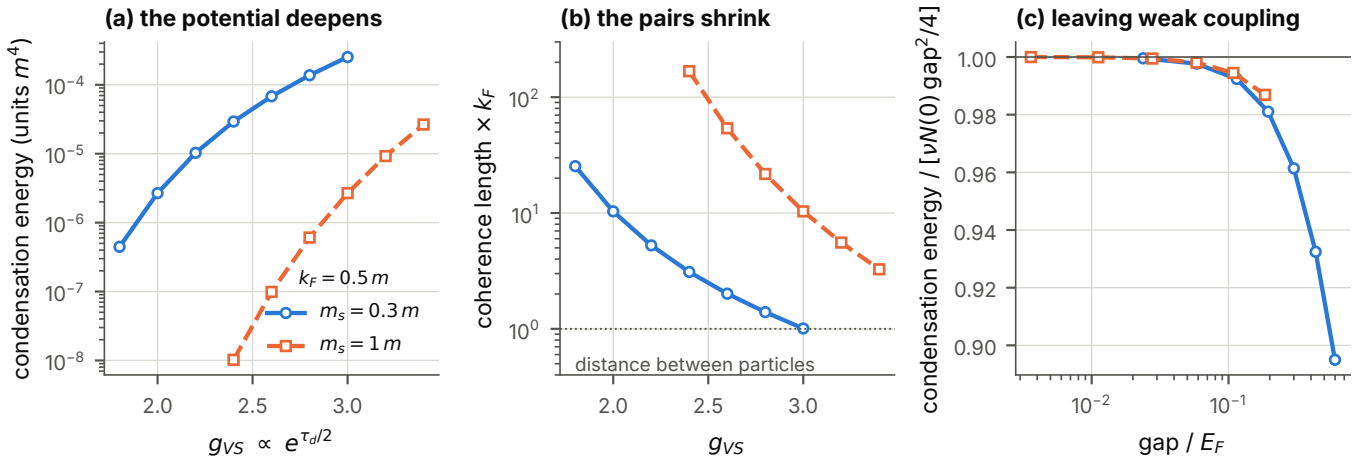


FIG. 10. The paired state along the clock, from the CJT potential, for the two sets of Table VI. (a) The condensation energy against the coupling of the moment. (b) The coherence length of the pair in units of $1/k_F$. (c) The condensation energy divided by its weak-coupling value, against the gap in units of the Fermi kinetic energy: the departure from weak coupling.

particle falls monotonically along the clock (Table VII), from $-0.0192\,m$ at $C = 4$ to $-0.405\,m$ at $C = 29.6$. At every one of these moments the matter of lowest energy is in the abnormal state. The exchange energies of the two modes, which the mean field omits, are 0.44 and 0.51 per cent of the mean-field energy at the first moment and 0.90 and 1.22 per cent at the last (for $m_s/m = 0.3$): the mean field remains a good approximation along the whole history. A gas of the fermions that is dilute and unbound at early times thus has, at late times, a state of lower energy: drops of dense abnormal matter, bound by up to 40 per cent of the rest mass at the last moment of the table. Whether and how fast a gas reaches that state is a question of kinetics that is not answered here.

D. Temperature

At high temperature the expectation value of the Vorton-Scalar vanishes and the fermions are massless. The temperature $T_{1/2}$ at which the expectation value has fallen to one half is of the order of the four-dimensional vacuum value $v_{VS} = m/g_{VS}$, which falls as $e^{-\tau_d/2}$ while the masses stay fixed. In units of the constant fermion mass, $T_{1/2}$ falls monotonically as the coupling grows, for each of the three ratios $m_s/m = 0.3$, 1 and 3 examined; for $m_s/m = 1$, from $2.33\,m$ at $g_{VS} = 0.3$ to $0.37\,m$ at $g_{VS} = 4$ [Fig. 11(d)]. *Matter of a fixed temperature melts at a definite moment of the deflation.* For $m_s/m = 0.3$ no melting temperature can be given from $C = 32$ on: there the one-loop potential of the vacuum turns over so close to the vacuum value that the Hartree potential ceases to exist before the expectation value has fallen,

TABLE VII. Self-bound matter along the clock, in the mean-field approximation without the negative-energy states, for $C = 4\,e^{\tau_d}$: the lowest energy per particle minus the fermion mass, and the Fermi momentum at which it is reached.

τ_d	C	$(E/N - m)/m$	k_F/m
0.00	4.00	−0.0192	0.981
0.25	5.14	−0.0787	0.921
0.50	6.59	−0.1345	0.866
0.75	8.47	−0.1869	0.813
1.00	10.87	−0.2362	0.764
1.25	13.96	−0.2825	0.718
1.50	17.93	−0.3259	0.674
1.75	23.02	−0.3668	0.633
2.00	29.56	−0.4051	0.595

and the program marks these moments as not defined.

E. The vacuum

The one-loop potential of the Vorton-Scalar in vacuum contains the negative contribution of the fermion loop [20], which grows with the fourth power of the field and eventually wins: beyond some field the potential lies below its value in the vacuum. The larger C, the smaller that field: 24.7 times the vacuum value at $C = 1$, and 1.77 times at $C = 10$ [Fig. 11(c)]. As the deflation proceeds, the barrier between the vacuum of the model and the region where the one-loop potential is unbounded below moves in toward the vacuum.

The one-loop potential is not reliable at fields far from the vacuum value, and a potential that falls at large field signals that the effective theory needs completion there, not that the vacuum decays. What the result establishes is a trend: every statement of the model that rests on the vacuum of the Vorton-Scalar becomes less secure with time. The order of this event relative to the binding of the pair was given in Sec. VII.

A pair condensate in the vacuum. The earlier paper found from the ladder equation with a momentum cutoff that the pair can condense without a medium only if $\alpha_{\rm VS}$ exceeds a critical value α_c of order one [21]; with the cutoff $100\,m$ the critical values lie between 2.13 and 2.93 for the six mass ratios of Table IV. On the deflating spacetime $\alpha_{\rm VS}$ reaches such a value at a definite moment. But at that moment the coupling is strong, the exchange of one boson is no longer a controlled approximation, and the vacuum of the model is no longer protected. A pair condensate in the vacuum is not reached by the deflation within the range of validity of the model.

TABLE VIII. The potential V of the paired state for six states of the order parameter, in units of $N(0)\Delta_F^2$, where Δ_F is the gap of a unitary state. Each state is taken at the magnitude of the order parameter at which its potential is stationary. d_-^2 and d_+^2: the squares of the two fourfold gaps of the state, in units of Δ_F^2. Flat and positive: the numbers of eigenvalues of the matrix of second derivatives of the potential that vanish and that are positive.

state	d_-^2	d_+^2	V	flat	positive
pair along the frame axis of $x0$ (hidden-axis singlet)	1.000	1.000	−2.0000	4	4
pair along the frame axis of $x5$	1.000	1.000	−2.0000	4	4
pair along the frame axis of $x7$	1.000	1.000	−2.0000	4	4
unitary, rotated between $x0$ and $x5$	1.000	1.000	−2.0000	4	4
unitary, general direction and phase	1.000	1.000	−2.0000	4	4
half-paired: $\eta_{ab}\Delta^a\Delta^b = 0$	0.000	1.000	−1.0000	—	—

X. ORDER PARAMETER, HIDDEN SYMMETRY AND VORTICES

A. Frame components and coordinate components

The order parameter of the superfluid is the expectation value Δ^a of the vector pair operator (36) along the extra axis $a \in \{0, 5, 6, 7\}$: four complex numbers. On the deflating spacetime the index a is a *frame* index: Δ^a is the component along the unit vector e_a. The coordinate components along the deflating coordinates are $\Delta_t = b_t(x4)\,\Delta^t$, and they shrink with the scale factors when the frame components are constant. All statements below are about frame components.

The energy of the paired state depends on the order parameter through the 8×8 gap matrix M on the internal states of a pair. The identity of the earlier paper,

$$\tfrac{1}{8}\,\mathrm{Tr}\big[(\mathsf{M}\mathsf{M}^\dagger)^2\big] = 2\Big(\sum_a |\Delta^a|^2\Big)^2 - \Big|\sum_{a,b}\eta_{ab}\,\Delta^a\Delta^b\Big|^2, \quad (46)$$

holds on the deflating spacetime with the frame metric η_{ab} of the four axes, to 2×10^{-14}. In coordinate components the same expression contains the metric g_{tt} of the moment.

B. The ground state: no direction is preferred

Table VIII gives the potential of the paired state for six states of the order parameter. Every *unitary* state, in which Δ^a is a real unit vector times a phase, has all eight gaps equal and the lowest potential, whichever direction the vector has. It has four flat directions, the Goldstone

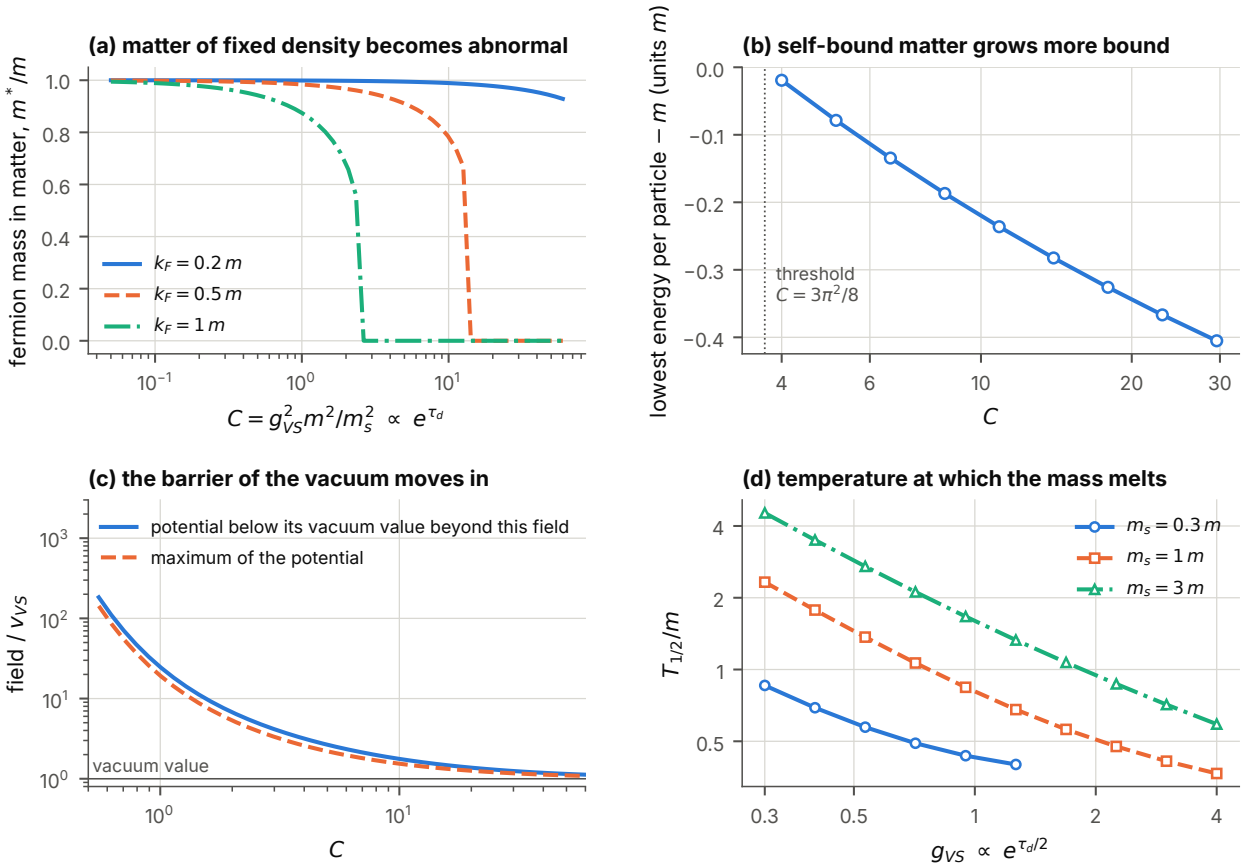


FIG. 11. Matter, vacuum and temperature along the clock. (a) The fermion mass in matter of three fixed densities against $C \propto e^{\tau_d}$: it drops to zero at the critical value. (b) The lowest energy per particle of self-bound matter, minus m. (c) The field at which the one-loop potential of the vacuum has its maximum, and the field beyond which it lies below the vacuum value, in units of the vacuum value. (d) The temperature at which the fermions have lost half of their mass, against the coupling, for three ratios m_s/m.

modes (one phase and three rotations of the direction), and four directions of positive curvature. The *half-paired* states, in which four of the eight gaps vanish, have half the condensation energy.

One might have expected the deflation to select the direction of the order parameter: the hidden axis $x0$, which does not deflate, against the three axes that do. *In the sector of fields that do not depend on the extra coordinates it does not.* The reason is the central observation of Sec. III: in this sector the weighted field obeys the flat equations, and their internal symmetry rotates the four frame axes into one another. The symmetry of the manifold of ground states is larger than the symmetry of the spacetime.

C. The internal symmetry of the sector, and the symmetries of the spacetime

For fields that depend on $x1, x2, x3, x4$ only, the equation of the weighted field contains five matrices: $\alpha^1, \alpha^2, \alpha^3$, β and $i\beta\chi$, the last through the coupling of the phase mode. The matrices that commute with all five generate the *internal symmetry* of the sector. The program computes it again for the deflating spacetime, item by item, and finds it unchanged:

- the algebra is $su(2)_A \oplus su(2)_B \oplus u(1)_F \oplus u(1)_E$, of dimension 8;
- the eight internal states of a fermion form the two doublets $(2,1) + (1,2)$;
- the four vector pair operators (36) form the $(2,2)$, with fermion number 2, and their one invariant quadratic form is the frame metric of the four axes, with the signs $(+,+,+,+)$ up to an overall sign;
- a condensate along one axis breaks $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$ to $SO(3)$.

Here $U(1)_E$ is a second phase symmetry of the sector, with a generator P that has the square one and distinguishes the doublets of $SU(2)_A$ from those of $SU(2)_B$.

Six of the generators are the products $\gamma^A\gamma^B$ of the six pairs taken from the four axes $x0, x5, x6, x7$. On the flat

7+1 spacetime they are the rotations of the four extra space-like coordinates, and the internal symmetry $SO(4)$ is a symmetry of the spacetime. On the deflating spacetime this is no longer so (Fig. 1). Of the six, the spacetime keeps

- none, if the three rates differ;
- one, the rotation of the two coordinates with the same rate, if two rates are equal;
- three, the rotations of $x5, x6, x7$ among themselves, if the three rates are equal.

The three rotations that mix the hidden axis $x0$ with a deflating coordinate are never symmetries of the deflating spacetime: $x0$ keeps its size and the other three shrink. The generators that are lost as symmetries of the spacetime remain exact symmetries of the sector. They are *hidden symmetries*: symmetries of the equations of a sector that are not transformations of the spacetime. The companion paper shows that on the 4+4 spacetime this was the situation from the start [1, 3].

D. Vortices and their zero modes

Because the vacuum value of the Vorton-Scalar breaks the phase symmetry $U(1)_{\rm VS}$, the field has line-like defects [22]: along a straight line, say the $x3$ axis, its magnitude vanishes, and around the line its phase winds by 2π times an integer. For unit winding, $\Phi_{\rm VS}$ is $v_{\rm VS}/\sqrt{2}$ times a profile that rises from 0 on the line to 1 at a distance of the order of $1/m_s$, times $e^{i\theta_{12}}$ with the angle θ_{12} around the line in the plane of $x1$ and $x2$.

Energy per unit length. It is proportional to the square of the four-dimensional vacuum value and so falls as $e^{-\tau_d}$: in the four-dimensional description vortex lines become lighter as the deflation proceeds, while their core, of the radius $1/m_s$, keeps its size.

Fermions on the line. On the line the fermion mass vanishes, and fermions can be trapped there [23]. In the field of the vortex the Hamiltonian of a fermion is

$$\sum_{j=1}^{3} \alpha^j p_j + m\,(\text{profile})\,\beta\left(\cos\theta_{12} + i\chi\,\sin\theta_{12}\right), \qquad (47)$$

with the momenta p_j along $x1, x2, x3$. A state of zero energy at zero momentum along the line is a constant spinor w_z times a function that decays away from the line. Inserting this form, the Hamiltonian annihilates the state for every angle if and only if

$$\left(i\alpha^1 + \beta\right) w_z = 0, \qquad \left(i\alpha^2 + i\beta\chi\right) w_z = 0. \qquad (48)$$

The five matrices $\alpha^1, \alpha^2, \alpha^3, \beta, i\beta\chi$ are Hermitian, have the square one and anticommute with one another. The two conditions state that w_z is an eigenvector, with the eigenvalue -1, of the two commuting matrices $i\beta\alpha^1$ and $\alpha^2\beta\chi$, and each joint eigenspace of these two matrices has the dimension $16/4 = 4$. *A vortex line of unit winding carries four zero modes of the sixteen-component field.*

For a momentum p_3 along the line the energy is the eigenvalue of $\alpha^3 p_3$. On the four zero modes α^3 equals the matrix

$$P_c = i\,\alpha^1\alpha^2\alpha^3\chi, \qquad (49)$$

the product of the five matrices, which commutes with each of them, is Hermitian and traceless and has the square one; it is, up to its sign, the generator P of $U(1)_E$. Its trace on the four zero modes vanishes. *Two zero modes move along the line in one direction at the speed of light and two in the other*, and the direction of motion is tied to the charge under $U(1)_E$. For winding -1 the two directions are exchanged.

Equations (48) contain only the matrices of the frame and the fermion mass, neither of which changes. The four zero modes therefore exist at every moment of the deflation. The count also holds on flat spacetime, where it corrects the earlier paper [2]: that paper quoted the count for a four-component field, one zero mode for each unit of winding, whereas the sixteen components are equivalent to four such fields.

Massless fermions that run along a vortex line carry a current [24]; a closed loop of the line can be held open by that current, and such loops are the *vortons* of the literature on cosmic strings [25]. Because the energy per length of the line falls as $e^{-\tau_d}$ while the fermions on it stay massless, the balance between the tension of a loop and the current on it shifts in favor of the current as the deflation proceeds. This paper establishes the ingredients of that statement; it does not compute the equilibrium radius of a loop.

XI. COSMOLOGY

Hypothesis H3 says that the superfluid explains the dark matter, or the Hubble tension, or both. The earlier paper examined it with the extra coordinates static. It found that the dark radiation of the model does not relieve the Hubble tension and that a superfluid of the fermions cannot be the dark matter of galaxies, although heavy bound pairs could be cold dark matter of an ordinary kind [2]. This section examines what changes when $x5, x6, x7$ deflate. The deflation enters cosmology in several ways, and each requires an assumption beyond the hypothesis of deflation. The assumptions are stated first and are kept apart throughout, so that the reader can see which conclusion rests on which.

A. Assumptions, and the equation of the expansion

A1: ordinary space expands. The coordinates $x1, x2, x3$ carry the scale factor $a_{\rm sp}(x4)$ of standard cosmology, with the Hubble rate $H = d\ln a_{\rm sp}/dx4$; the hid-

den coordinate $x0$ is static; $x5, x6, x7$ deflate at constant rates. The metric of Sec. II is the case $a_{\rm sp} = 1$.

A2: gravity is the Einstein theory in eight dimensions, with a constant G_8. The constant of gravitation that an observer in ordinary space measures is then G_8 divided by the proper volume of the extra coordinates,

$$G_N(x4) \;\propto\; \frac{1}{V_d} \;\propto\; e^{\mathrm{gsum}\, x4}, \qquad \frac{1}{G_N}\frac{dG_N}{dx4} = \mathrm{gsum}. \tag{50}$$

Newton's constant *grows* as the extra volume shrinks: gravity was weaker in the past. That a changing volume of extra dimensions makes the constants of four dimensions change, and that nucleosynthesis bounds the change, was pointed out long ago [26]; for a review of varying constants see Ref. [27]. By Sec. II C, A2 is an assumption that the deflating spacetime itself puts in doubt: without matter the metric solves the equations of Einstein-Gauss-Bonnet gravity, not Einstein's.

For a diagonal metric whose seven directions other than the time have the rates H, H, H (ordinary space), 0 (the coordinate $x0$) and $-\mathrm{g}55, -\mathrm{g}66, -\mathrm{g}77$, the component of Einstein's equations along the time is the sum of the products of all pairs of rates, set equal to $8\pi G_8$ times the energy density in eight dimensions. Dividing the density by the extra volume turns G_8 into G_N:

$$3H^2 - 3\,\mathrm{gsum}\,H + \mathrm{gprod} = 8\pi\, G_N(x4)\,\rho, \tag{51}$$

with the density ρ of radiation, matter and the cosmological term in ordinary space. Equation (51) is the only one of Einstein's equations that is used; the others would determine the stresses of the unknown source that keeps the three rates constant.

The calculations take three equal rates, so that $\mathrm{gprod} = \mathrm{gsum}^2/3$, and the densities of the standard model of cosmology [28] as the earlier paper used them: baryons $\omega_b = 0.02237$, cold dark matter $\omega_c = 0.1200$, and $H_0 = 67.36\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ without deflation. The cosmological term is chosen so that Eq. (51) holds today. Without deflation the program reproduces the two numbers of the earlier calculation on which this section rests: the angle θ_* under which the sound horizon at last scattering is seen, $100\,\theta_* = 1.03965$, and that sound horizon, $r_s = 144.449$ Mpc.

B. The Hubble constant inferred from the sound horizon

The microwave background fixes the angle θ_* to high precision. A model that changes the history of the expansion must keep that angle, and the Hubble constant that it then requires is the *inferred* Hubble constant. The tension is that the standard model infers $67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [28] while the local distance network measures 73.50 ± 0.81 [29]. Table IX gives the inferred value on the deflating spacetime under A1 and A2, at fixed θ_* and fixed densities of baryons and dark matter.

The inferred Hubble constant falls, with the slope $d\ln H_0/d(\mathrm{gsum}/H_0) = -0.911$ at small rates. The reason is this. By Eq. (50) gravity was weaker in the early universe; the universe expanded more slowly at a given density; sound had more time to travel, and the sound horizon is larger. To keep the angle, the distance to last scattering must be larger, which requires a smaller Hubble constant. The deflation moves the inferred value *away* from the local measurement [Fig. 12(a)]. The term gprod of Eq. (51) plays no part in this: it is of the second order in the rate, and it acts as a constant, which the cosmological term that is adjusted to today's expansion absorbs almost exactly.

C. How fast can the coordinates deflate today?

Table X and Fig. 13 collect the bounds.

Under A2, lunar laser ranging measures $(dG_N/dt)/G_N$ directly, and Eq. (50) turns its limit into $\mathrm{gsum} < 2.23 \times 10^{-13}$ per year, that is, $3.24 \times 10^{-3}\,H_0$. At this rate the deflation clock has advanced by $\mathrm{gsum}\,t_0 = 3.1 \times 10^{-3}$ over the age t_0 of the universe, and the inferred Hubble constant is lower by $0.20\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Under A2, nucleosynthesis bounds the ratio of Newton's constant then to its value today; for a rate that was constant since, $\mathrm{gsum} < 0.064\,H_0$. This is weaker than the first bound but does not assume that the rate today equals the rate in the past.

Under HV, nucleosynthesis bounds the change of the Fermi constant at fixed fermion masses to 10 to 20 per cent, which gives $\mathrm{gsum} < 0.111$ to $0.234\,H_0$. HV is the assumption that the Vorton-Scalar itself plays the role of the Higgs field, so that $G_{\rm Fermi} = 1/(\sqrt{2}\,v_{\rm VS}^2)$ grows as e^{τ_d} while every mass stays constant. It is *not* the hypothesis H2, and the earlier paper already noted that the Vorton-Scalar, a singlet of all non-Abelian symmetries, cannot be the Higgs field of the Standard Model; the bound is given only to show what the assumption would imply.

Under the hypothesis H2, atomic clocks give by far the strongest bound, and one that needs no assumption about gravity. Section XII derives it: $\mathrm{gsum} < 1.16 \times 10^{-6}\,H_0/n_{\rm EW}$ with an exponent $n_{\rm EW}$ of order one or larger.

The fine-structure constant is bounded by the same clocks to parts in 10^{18} per year. This would bound the rate, to $4.6 \times 10^{-8}\,H_0$, only if an electromagnetic field propagated along the deflating coordinates. The model contains no electromagnetic field; the row shows how strongly such an extension would be constrained.

In the model as it stands, without H2, HV or an electromagnetic extension and without A2, no mass and no coupling of ordinary matter depends on the deflation, and none of these observations bounds the rate.

The consequence for the histories of Secs. VI to IX is severe. Under A2 the clock has advanced by at most 0.0031 during the whole of cosmic history, if the rate

TABLE IX. The Hubble constant inferred from the angular scale of the sound horizon if Newton's constant grows with the deflation (assumptions A1 and A2). G_N ratio: Newton's constant in the early universe divided by its value today, $e^{-\mathrm{gsum}\, t_0}$ with the age t_0 of the model. The rows at $\mathrm{gsum}/H_0 = 0.00324$ and 0.0644 are the bounds of lunar laser ranging and of nucleosynthesis (Table X).

gsum/H_0	gsum (per year)	H_0 ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	change	G_N ratio	r_s (Mpc)	t_0 (Gyr)
0	0	67.36	0.00	1.0000	144.45	13.81
0.00100	6.89×10^{-14}	67.30	−0.06	0.9990	144.52	13.82
0.00300	2.07×10^{-13}	67.18	−0.18	0.9971	144.66	13.84
0.00324	2.23×10^{-13}	67.16	−0.20	0.9969	144.67	13.84
0.0100	6.89×10^{-13}	66.74	−0.62	0.9905	145.14	13.89
0.0300	2.07×10^{-12}	65.48	−1.88	0.9714	146.56	14.05
0.0644	4.43×10^{-12}	63.23	−4.13	0.9384	149.12	14.34
0.100	6.89×10^{-12}	60.77	−6.59	0.9039	151.93	14.66
0.300	2.07×10^{-11}	43.41	−23.95	0.7024	172.36	17.10

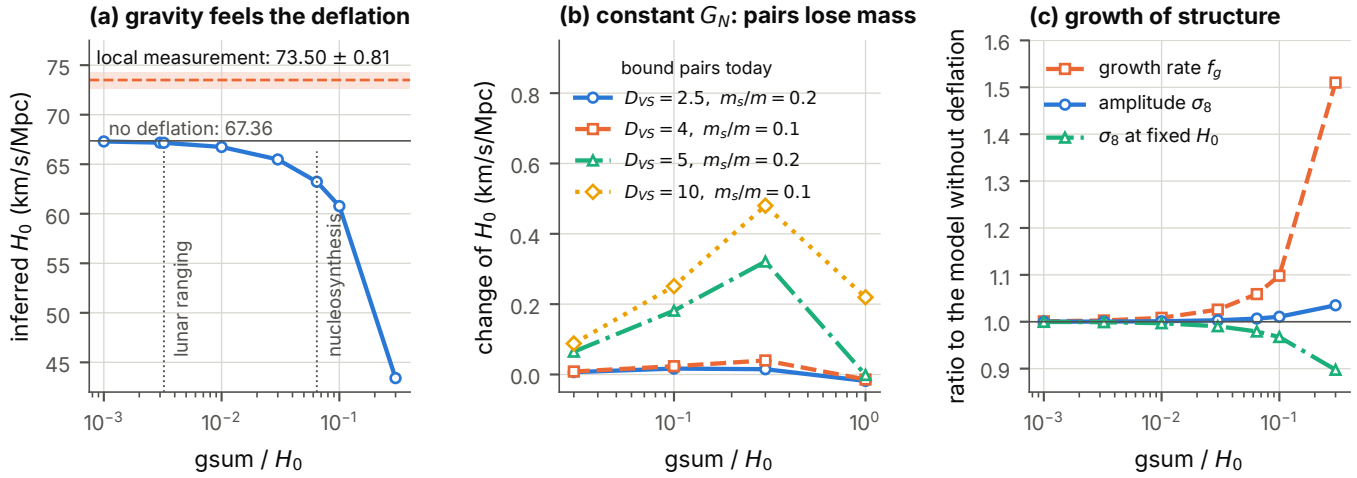


FIG. 12. The Hubble constant and the growth of structure on a deflating spacetime. (a) The Hubble constant inferred from the angular scale of the sound horizon against the rate of deflation, if gravity feels the deflation (A1, A2); the bounds of Table X and the local measurement are marked. (b) The change of the inferred Hubble constant if Newton's constant is held fixed and bound pairs lose rest mass. (c) Amplitude and rate of the growth of structure relative to the model without deflation.

TABLE X. Upper bounds on the present total rate of deflation, each under the assumption named. Uncertainties of the measured values are at 68 per cent unless stated; the bounds are at two standard deviations. The bound in units of H_0 uses $H_0 = 67.36\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. A2: gravity is the Einstein theory in eight dimensions. HV: the vacuum value of the Vorton-Scalar sets the electroweak scale. H2: the superfluid state is the Higgs field; the bound is given for $n_{\mathrm{EW}} = 1$ and scales as $1/n_{\mathrm{EW}}$ (Sec. XII).

observation	measured	assumption	bound on gsum (per year)	in units of H_0
lunar laser ranging [30]	$(dG_N/dt)/G_N = (7.1 \pm 7.6) \times 10^{-14}$ per year	A2	2.230×10^{-13}	3.237×10^{-3}
nucleosynthesis, Newton's constant [31]	G_N then, divided by G_N today: $0.99^{+0.06}_{-0.05}$ (two standard deviations)	A2; constant rate	4.434×10^{-12}	0.0644
nucleosynthesis, Fermi constant [32]	G_{Fermi} at $t = 1$ s within 20 (10) per cent of today's value, at fixed fermion masses	HV; constant rate	1.615×10^{-11} (7.627×10^{-12})	0.234 (0.111)
atomic clocks, mass ratio [33]	$(d/dt)\ln(m_p/m_e) = (-8 \pm 36) \times 10^{-18}$ per year	H2, $n_{\mathrm{EW}} = 1$	8.000×10^{-17}	1.161×10^{-6}
atomic clocks, fine-structure constant [33]	$(d\alpha_{\mathrm{em}}/dt)/\alpha_{\mathrm{em}} = (1.0 \pm 1.1) \times 10^{-18}$ per year	an electromagnetic field along $x5, x6, x7$ (not part of the model)	3.200×10^{-18}	4.645×10^{-8}

was constant; under H2 by at most $1.1 \times 10^{-6}/n_{\rm EW}$. A history in which pairs become bound, matter loses its fermion mass or the gap grows by orders of magnitude within the age of the universe requires that gravity does not feel the deflation, that the superfluid is not the Higgs field, or that the rate was much larger in the past than it is today.

D. Bound pairs lose rest mass

If the dark matter consists of bound pairs of the fermions, its particles have the rest mass $2m$ minus the binding energy. The binding energy grows with the coupling, so that the pairs become lighter as the deflation proceeds, and the density of dark matter falls slightly faster than the inverse volume. A dark matter that was heavier in the past raises the inferred Hubble constant. The program computes the effect for four examples with $\alpha_{\rm VS}$ between 0.4 and 1 today, where the binding energy is at most 8.2 per cent of the rest mass [Fig. 12(b)]. If gravity feels the deflation, the inferred Hubble constant changes by between -4.11 and -0.19 km s^{-1} Mpc^{-1} at the rates that nucleosynthesis and lunar laser ranging allow: the weaker gravity of the past outweighs the loss of mass. If Newton's constant is held fixed and the deflation acts on the dark sector alone, rates between 0.03 and $1\,H_0$ raise the inferred Hubble constant by at most $+0.48$ km s^{-1} Mpc^{-1}, for a loss of up to 7.7 per cent of the rest mass. This is the only effect of the deflation found here that raises the inferred Hubble constant, and in these examples it supplies less than a tenth of the 6.1 km s^{-1} Mpc^{-1} that separate the model from the local measurement.

E. The growth of structure

Under A1 and A2 the density contrast δ_m of matter obeys, in the Newtonian limit, the usual equation with the Newton's constant of the moment,

$$\frac{d^2\delta_m}{dx4^2} + 2H\,\frac{d\delta_m}{dx4} = 4\pi\,G_N(x4)\,\rho_m\,\delta_m, \qquad (52)$$

with the density ρ_m of matter. Two effects oppose each other [Fig. 12(c)]. At a *fixed* Hubble constant, weaker gravity in the past lowers the amplitude σ_8 of clustering today: $d\ln\sigma_8/d(\mathrm{gsum}/H_0) = -0.317$. But the Hubble constant that keeps the angle of the sound horizon is lower, the cosmological term takes over later, and matter clusters for longer; with that Hubble constant the net change is $+0.102$ per unit of gsum/H_0. The amplitude therefore barely moves, by $+0.03$ per cent at the bound of lunar laser ranging and $+0.67$ per cent at the bound of nucleosynthesis, while the growth rate today, $f_g = d\ln\delta_m/d\ln a_{\rm sp}$, rises by the factors 1.0027 and 1.0589. A deflating universe would show itself in the product $f_g\sigma_8$, which redshift-space distortions measure, rather than in σ_8.

F. Dark matter

What does not change. The estimates of the earlier paper that decide whether the particles of the model can be degenerate or condensed in observed halos contain the masses, the densities and the velocities only, and the masses do not change along the deflation. Computed again, every number agrees: fermions with eight internal states are degenerate in the core of a dwarf galaxy only if they are lighter than 0.276 keV, and bound pairs are condensed there only if they are lighter than 262 eV. The verdict of the earlier paper on these two possibilities stands.

Self-interaction, and the epoch of the resonance. The cross-section per mass of bound pairs must respect the bound of about 1 cm^2/g from colliding clusters [34, 35], which the earlier paper turned into a smallest allowed fermion mass: 918 MeV for $\alpha_{\rm VS} = 0.1$. For a given particle the masses are constant and so is this cross-section. What the deflation changes is whether the pair *exists*: it is bound only after $D_{\rm VS}$ has grown to D_c. And at that moment the scattering length of two unbound fermions diverges (Sec. VII), so that their cross-section rises to the limit that unitarity allows, which is larger the slower the particles. For the example particle, $m = 918$ MeV and $m_s = 54.7$ MeV, at the relative velocities 10, 156 and 1000 km/s [Fig. 8(c)]: one unit of the clock before the binding the cross-section per mass is 0.84 cm^2/g at all three velocities; at the binding it peaks at 1.3×10^7, 5.2×10^4 and 1.3×10^3 cm^2/g; and it exceeds 100 cm^2/g during an interval of the clock of 0.27, 0.27 and 0.26 around the binding. *A deflating universe whose dark matter consists of these fermions passes through an epoch of strong, velocity-dependent self-interaction*, strongest in the smallest halos. Whether the epoch lies in the observable past depends on the rate. Under A2 and a constant rate it does so only if $D_{\rm VS}$ today exceeds D_c by less than a third of a per cent, a coincidence that nothing in the model explains.

How fast the quantities drift. With $n = d\ln(\mathrm{quantity})/d\tau_d$, the relative change of a quantity over the age of the universe is $n\,\mathrm{gsum}\,t_0$. The couplings themselves drift with $n = 1$ or $\frac{1}{2}$ (Table I). Quantities near a threshold drift much faster: the scattering length just below the binding, at $D_{\rm VS} = 1.5$, with $n = 9.1$; the binding energy just above it, at $D_{\rm VS} = 1.8$, with $n = 30$; the gap of a weakly paired state with $n = 1/\lambda$. These are the quantities in which a slow deflation would first be seen.

G. The quanta that the deflation creates

The temperature (25) of the quanta that the deflation creates in modes that depend on a deflating coordinate is,

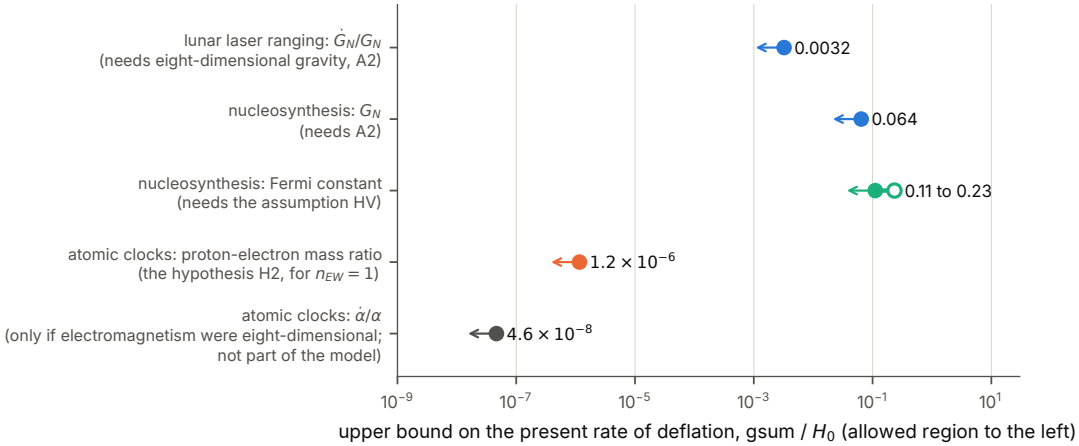


FIG. 13. Upper bounds on the present total rate of deflation in units of the Hubble constant, each under the assumption of Table X. The allowed region is to the left of each bound.

for three equal rates at the bound of lunar laser ranging, 2.9×10^{-33} K. The number of created particles of mass m per mode is $e^{-2\pi m/\mathrm{rate}_t}$, with an exponent above 10^{37} for every mass above 1 keV at any rate allowed today. *No particle is created today.* Creation by deflation matters only for an epoch in which the rate was comparable with the masses, which no bound of this section addresses.

H. Verdict on hypothesis H3

The Hubble tension. The mechanism of the earlier paper, the phase mode of the Vorton-Scalar as dark radiation, is untouched at the allowed rates, since the couplings drift by parts in a thousand; its verdict stands and is not improved. The deflation itself contributes two effects. The first, through Newton's constant, has the wrong sign and is bounded to 0.2 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The second, the loss of mass of bound pairs, has the right sign and is bounded, in the examples computed, to 0.5 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. *The hypothesis that the model resolves the Hubble tension is not supported by the deflation, and under A2 it is mildly disfavored.*

Dark matter. The conclusions of the earlier paper about the mass ranges hold unchanged. The deflation adds one qualitative prediction, the epoch of the resonance, and one quantitative one, the shift of the growth rate. Neither is in conflict with observation at the allowed rates, and neither is detectable at the rate that lunar laser ranging allows. *The hypothesis that the superfluid is the dark matter is neither strengthened nor weakened by the deflation.* It acquires a history, whose observable part is empty if gravity is eight-dimensional and the rate constant.

What was not computed. The earlier paper computed the temperature and polarization spectra of the microwave background with its own Einstein-Boltzmann program and forecast the constraints from them. That calculation has not been repeated with a Newton's constant that depends on time, for two reasons. First, the equations of the perturbations contain the stresses of the source that keeps the rates of deflation constant, which the hypothesis does not specify; a spectrum can be computed only under a further assumption about that source or about the theory of gravity. Second, it requires a version of the program in which the gravitational coupling of the background and of the perturbations depends on time while the rates of Thomson scattering and of recombination do not, validated again against the constant case. What is established here is the effect on the *background*. At the bound of lunar laser ranging Newton's constant at last scattering was 0.9969 of today's value; the effect on the spectra beyond the shift of the angle of the sound horizon is of that relative size and has not been computed.

XII. THE HIGGS HYPOTHESIS

A. The hypothesis and the conditions

Hypothesis H2 says that the superfluid state, the condensate of the fermion pair along the hidden axis, *is the Higgs field* [36]. In the Standard Model the Higgs field is a doublet of the weak-isospin group with a vacuum value $v_{\mathrm{EW}} = 246$ GeV, which breaks the electroweak symmetry, gives the W and Z bosons and the quarks and charged leptons their masses, and leaves one scalar particle of 125 GeV. Its four real components form a vector of a group $SO(4)$, which the vacuum value breaks to $SO(3)$ [37]. The earlier paper derived from this seven conditions that a condensate must meet in order to *be* the Higgs field (Table XI) and found the hypothesis unproven on the flat 7+1 spacetime, with one structural point in its favor: the order parameter has the representation and the breaking pattern of the Higgs field under a symmetry of the spacetime. This section repeats the

test on the deflating spacetime.

Two statements must be kept apart. H2 is the hypothesis that the *pair condensate* is the Higgs field; the electroweak scale is then the scale of the condensate. HV, used in one row of Table X, is the different assumption that the vacuum value of the *Vorton-Scalar* sets the electroweak scale. HV is not examined here.

B. What the deflation leaves unchanged

The pair operators, their transformation properties and the internal symmetry are properties of the Dirac matrices of the frame, which the deflation does not change, and of the equations of the fields that do not depend on the extra coordinates, which for the weighted fields are the flat equations. By Sec. X C the four vector pair operators form the $(2,2)$ of $SU(2)_A \times SU(2)_B$, the representation of a Higgs field, and a condensate along one axis breaks $SO(4)$ to $SO(3)$. Hence, *within the sector*, C1 holds as on flat spacetime: the pair along the hidden axis is invariant under the rotations of $x1, x2, x3$, which remain symmetries of the spacetime, and under the boosts of ordinary spacetime as they act on the fields of the sector. (The boosts are no longer symmetries of the deflating spacetime as a whole; in the sector this shows only in the dependence of the couplings on the time.) C3 and C4 hold in the sector as well, in the sense that the pattern is right under its internal symmetry.

C. What the deflation changes

On the flat 7+1 spacetime the six generators of $SO(4)$ are rotations of the extra dimensions. This is what the earlier paper counted in favor of the hypothesis: a symmetry of extra dimensions is the kind of symmetry from which a gauge interaction can arise, as in the theories of Kaluza and Klein [38]. By Sec. X C the deflating spacetime keeps none, one or three of the six, and never one of the three that mix the hidden axis with a deflating coordinate.

For equal rates the part of $SO(4)$ that remains a symmetry of the *spacetime* is $SO(3)$, the rotations of the three deflating coordinates: precisely the subgroup that a condensate along $x0$ would leave unbroken. The deflation thus breaks $SO(4)$ to $SO(3)$ *explicitly*, at the level of the geometry, in the same pattern in which the hypothesis needs the condensate to break it *spontaneously*. Three consequences follow. First, the three broken generators, which under H2 would correspond to the massive weak bosons, are not symmetries of the spacetime at all; a gauge interaction for them cannot arise from the geometry of the deflating spacetime. Second, in the sector the full $SO(4)$ nevertheless acts as an exact symmetry, but as a hidden one. The companion paper finds exactly this situation on the 4+4 spacetime, flat or deflating, where the hypothesis has no geometric basis [3]. *The deflation puts the 7+1 model in the position of the 4+4 model.* Third, the potential of the order parameter does not single out the hidden axis (Sec. X): within the sector the direction of the condensate is as free as on flat spacetime. Outside the sector the axes differ, because a variation of the order parameter along a deflating coordinate costs a gradient energy that grows as $1/b_i^2$; the hidden symmetry is exact only for fields that are strictly independent of the extra coordinates.

D. Condition C2: a condensate in empty space

The earlier paper left C2 open: a pair condensate forms in the vacuum only if $\alpha_{\rm VS}$ exceeds a critical value α_c of about 2 or more, depending on the cutoff. On a deflating spacetime $\alpha_{\rm VS}$ grows as e^{τ_d} and reaches any such value at a definite time. Section IX showed what else has happened by then: the exchange of one boson is no longer a controlled approximation, and the field at which the one-loop potential of the Vorton-Scalar falls below its vacuum value has moved to within a factor of two of the vacuum value. *C2 can be reached by waiting, but not within the range of validity of the model.* Its status remains open.

E. A new condition: the electroweak scale must not drift

Suppose that H2 holds. The electroweak scale $v_{\rm EW}$ is then the scale of the pair condensate in the vacuum, as in models in which a condensate of fermions replaces the elementary Higgs field [39]. That scale is determined by the parameters of the model: the masses m and m_s, the cutoff, and the coupling $\alpha_{\rm VS}$. On the deflating spacetime the masses are constant and $\alpha_{\rm VS} \propto e^{\tau_d}$. Therefore

$$v_{\rm EW} \;\propto\; e^{\,n_{\rm EW}\,\tau_d}, \qquad n_{\rm EW} = \frac{d\ln v_{\rm EW}}{d\ln \alpha_{\rm VS}}. \tag{53}$$

The exponent cannot vanish: below the critical coupling there is no condensate, so that the scale depends on the coupling, and most strongly just above the critical coupling.

Under H2 the masses of the electron and of the quarks are proportional to $v_{\rm EW}$ and change at the rate $n_{\rm EW}$ gsum. The mass of the proton is due mostly to the strong interaction and does not follow. The ratio m_p/m_e of the proton mass to the electron mass therefore changes at the rate $n_{\rm EW}$ gsum. Comparisons of atomic clocks find this ratio constant to 8.0×10^{-17} per year at two standard deviations [33]. Hence

$$\text{gsum} < \frac{8.0\times10^{-17}\ \text{per year}}{n_{\rm EW}} = \frac{1.16\times10^{-6}\,H_0}{n_{\rm EW}}. \tag{54}$$

The bound needs no assumption about gravity. For $n_{\rm EW} = 1$ it is 2787 times stronger than the bound that lunar laser ranging gives if gravity is eight-dimensional.

TABLE XI. The conditions that a condensate must meet in order to be the Higgs field, and their status on the deflating 7+1 spacetime. The last row is new.

condition	status
C1: a scalar under the Lorentz group of ordinary spacetime	met in the sector of fields that do not depend on the extra coordinates
C2: non-zero in empty space	open; reached by the deflation only outside the range of validity of the model
C3: a doublet of the weak-isospin group with a neutral component	only under a hidden symmetry (three different rates), or with $SO(3)$ alone as a symmetry of the spacetime (equal rates)
C4: breaks $SO(4)$ to $SO(3)$	met in the sector; the spacetime itself has at most the $SO(3)$
C5: scale 246 GeV, with a scalar particle of 125 GeV with the observed couplings	unfavorable (estimate of the earlier paper, unchanged)
C6: its couplings give the quarks and leptons their masses	not addressed
C7: no further light or charged particles that would have been seen	additional particles are predicted
new: the electroweak scale does not drift	requires gsum $< 1.16 \times 10^{-6}\, H_0/n_{\mathrm{EW}}$

The size of the exponent. For the model itself, with massive fermions and a massive radial mode, the scale of the vacuum condensate has not been computed as a function of the coupling. It is known in closed form for the case on which the earlier paper tested its ladder equation: massless fermions and massless bosons between an infrared scale and a cutoff [21]. There the scale of the condensate obeys

$$\varrho_c \ln\frac{\text{cutoff}}{\text{scale}} = \arctan\frac{1}{2\varrho_c}, \qquad \varrho_c^2 = \frac{1}{4}\Big(\frac{\alpha_{\mathrm{VS}}}{\alpha_c} - 1\Big), \quad (55)$$

where ϱ_c is only an abbreviation for the square root that the second equation defines, from which

$$n_{\mathrm{EW}} = \frac{(1+4\varrho_c^2)\,\arctan(1/2\varrho_c)}{8\,\varrho_c^3} + \frac{1}{4\varrho_c^2}. \quad (56)$$

The program verifies Eq. (56) against the numerical derivative to 3×10^{-8} and tabulates it with the bound (54) (Table XII). The exponent diverges at the critical coupling, equals $1 + \pi/2 = 2.57$ at twice the critical coupling, and exceeds 1 for $\alpha_{\mathrm{VS}} < 3.27\,\alpha_c$. It falls below 1 only where the scale of the condensate approaches the cutoff, that is, where the four-dimensional description ends. Within the range in which a condensate of the kind that H2 needs can be described at all, n_{EW} is of order one or larger.

The conclusion. Under H2 the deflation clock can have advanced by at most $\mathrm{gsum}\, t_0 < 1.1 \times 10^{-6}/n_{\mathrm{EW}}$ during the age of the universe at a constant rate. Every effect of the deflation described in Secs. VI to XI is then unobservably small. Conversely, a deflation at any rate at which those effects matter excludes H2. *The hypothesis H2 and a deflation of any consequence today exclude each other.* This conclusion does not depend on the unresolved conditions C2 and C5 to C7. It depends on two properties only: that the electroweak scale is set by the dynamics of the model, which is the content of H2, and that the coupling of the model follows the volume of the deflating coordinates, which is exact (Sec. V).

TABLE XII. The exponent n_{EW} with which the scale of the vacuum condensate follows the coupling in the scale-invariant ladder equation, and the bound (54) on the present rate of deflation that follows from it under hypothesis H2.

$\alpha_{\mathrm{VS}}/\alpha_c$	ln(cutoff/scale)	n_{EW}	gsum (per year)	gsum/H_0
1.05	12.082	147	5.4×10^{-19}	7.9×10^{-9}
1.10	7.998	54.0	1.5×10^{-18}	2.2×10^{-8}
1.25	4.429	15.1	5.3×10^{-18}	7.7×10^{-8}
1.50	2.702	6.05	1.3×10^{-17}	1.9×10^{-7}
2.00	1.571	2.57	3.1×10^{-17}	4.5×10^{-7}
3.00	0.870	1.15	6.9×10^{-17}	1.0×10^{-6}
5.00	0.464	0.540	1.5×10^{-16}	2.2×10^{-6}
10.0	0.215	0.230	3.5×10^{-16}	5.0×10^{-6}

F. Verdict on hypothesis H2

On the deflating spacetime the hypothesis that the singlet superfluid is the Higgs field is unproven, with less in its favor than on the flat 7+1 spacetime, and it is incompatible with any deflation that is fast enough to matter today. Whether a mechanism exists that would make the electroweak scale insensitive to the coupling of the pair is not known; none is present in the model.

XIII. WHAT HAS BEEN ESTABLISHED, AND WHAT HAS NOT

The sections above reported results of very different standing: identities verified in exact arithmetic, numerical results with a stated accuracy, consequences of additional assumptions, and estimates. This section sorts them. An item is called established if a machine check stands behind it (Appendix C); statements of interpretation deserve the reader's independent scrutiny more than the numbers do.

A. Established

Geometry (Sec. II). The curvature and the Einstein tensor of the deflating spacetime in closed form, for arbitrary rates and constants. The Killing vectors, 14, 15 or 20 in number, and the completeness of the lists. That the spacetime is not a solution of Einstein's equations without matter, with or without a cosmological constant; that with three equal rates it is an exact solution of Einstein-Gauss-Bonnet gravity without matter, for $\Lambda_8 = 3\,\mathrm{g}55^2$ and $\alpha_{\mathrm{GB}} = -1/(4\,\mathrm{g}55^2)$; and that with unequal rates it is not. The unbounded blueshift of a classical particle that moves along a deflating coordinate, and the incompleteness of its geodesic.

The Dirac field (Secs. III, IV). The Dirac matrices in coordinate components, the spin connection, the covariant constancy of the matrices, and the curvature of the spinor derivative. The Dirac equation, the weighted field, and the statement that fields which do not depend on $x5, x6, x7$ obey the flat equation. The exact solutions when one rate governs, and the law of creation (24) with the temperature (25). All of this is confirmed numerically by two independent methods.

The model along the clock (Secs. V to IX). For fields that do not depend on the extra coordinates, the model of the earlier paper on the deflating spacetime is the same model with the couplings of the moment, exactly. The masses are constant; g_{VS} grows as $e^{\tau_d/2}$; α_{VS}, G_{VS}, D_{VS} and C grow as e^{τ_d}. The modes of the Vorton-Scalar grow with the weight only, and the potential between two fermions has the unchanged range and the strength of the moment. The exchange amplitudes are those of flat spacetime times the square of the coupling of the moment. In weak coupling the gap grows according to Eq. (39), and the critical temperature follows it. Two fermions become bound at a definite time, at a scattering resonance, with a binding energy that starts quadratically; the order of the other events depends on m_s/m alone. The paired state follows the deflation if and only if Ξ_d is below about 0.2 to 0.5; before that it lags and accumulates broken pairs. The solution of the gap equation is a minimum of the effective potential at every moment examined. Matter of fixed density loses its fermion mass at a critical value of C, is self-bound for $C > 3.7011$ and is more strongly bound later; the melting temperature of the fermion mass falls; the barrier of the one-loop potential of the vacuum moves toward the vacuum value.

Order parameter and symmetry (Sec. X). The potential of the order parameter prefers no direction among the four frame axes. The internal symmetry of the sector, the representation of the pair operators and the breaking pattern are unchanged by the deflation. Of the six generators that belong to the pairs of the four extra axes, the deflating spacetime keeps none, one or three as symmetries of the spacetime. A vortex line of unit winding carries four fermion zero modes, two for each direction along the line, at every moment; its energy per length falls as $e^{-\tau_d}$.

Cosmology (Sec. XI), each item under the assumption named there. The equation of the expansion (51) and the fall of the inferred Hubble constant, verified by an independent integration. The bounds of Table X, and that in the model as it stands the atomic clocks give none. The effect of the loss of mass of bound pairs. The growth of structure. The unchanged estimates of degeneracy and of condensation in halos, the epoch of the resonance, and the drift of the quantities. The temperature of the created quanta at the allowed rates.

The Higgs hypothesis (Sec. XII). Under H2 the electroweak scale follows the coupling with the exponent n_{EW}, and atomic clocks bound the rate of deflation by Eq. (54). The exponent is given in closed form for the scale-invariant ladder equation.

B. Not established

The source of the deflation. The metric is prescribed. For equal rates it solves the equations of Einstein-Gauss-Bonnet gravity without matter; whether that solution is stable has not been examined. For unequal rates no field equation has been found that the metric solves, and the stress tensor that Einstein's equations would require has not been identified with any matter. All results on fields are results on a given background.

The theory of gravity and the assumption A2. If gravity is of the Einstein-Gauss-Bonnet kind, the relation (50) between Newton's constant and the volume of the deflating coordinates, and with it every result of Sec. XI that is marked A2, would have to be derived anew. Only the time component of Einstein's equations was used; whether a source exists that keeps three rates constant while ordinary space expands has not been examined.

The reaction of the created particles on the metric. The particles created by the deflation gain energy without bound. Their stress must eventually change the metric; this has not been computed. Nor do the coordinates cover the whole future of particles that move along deflating coordinates; what lies beyond is outside the hypothesis.

The restriction to the sector. That the fields do not depend on $x5, x6, x7$ is assumed, as in the earlier paper. The deflation supports the assumption only if the deflating coordinates are periodic (Sec. IV).

Unequal rates with simultaneous crossings. No closed form is known for a mode that depends on two deflating coordinates whose crossings coincide; the results are numerical.

The hidden coordinate. That the fields do not depend on $x0$ is assumed, as in the earlier paper. Nothing in the deflation explains it.

The end of the history. The couplings grow without bound. The four-dimensional description with the exchange of one boson holds only while α_{VS} is not large, and the eight-dimensional theory itself is an effective theory, with couplings of negative mass dimension, that must fail when the proper size of the deflating coordinates reaches

its scale of validity. What happens then is outside the model. Every history of Secs. VI to IX ends in a regime that the two-loop truncation does not control; the numbers at the last moments of each table indicate the trend and are not predictions.

The energy. The energy of matter is not conserved while the couplings change. Where it goes, into whatever drives the deflation, is not described.

Dynamics. Sections VI, VII and IX give sequences of equilibrium states. Only for the paired state has the ability to follow been examined, in a model without dissipation; with dissipation the broken pairs would relax, and their lifetime is not known. The time-dependent two-body problem at the moment of binding, which decides how much of the population ends in the bound state, has not been solved. How a gas reaches the self-bound state, and whether the abnormal state forms by nucleation, are open. The equilibrium of a loop of vortex line with its current, and its stability, have not been computed.

The vacuum beyond one loop. The statements about the vacuum use the one-loop potential outside its range of reliability and draw only the conclusion stated in Sec. IX.

Cosmology. The spectra of the microwave background with a Newton's constant that depends on time have not been computed (Sec. XI). All bounds that reach back to nucleosynthesis assume a constant rate; a rate that was large early and is small today is neither excluded nor described. The consequences of the epoch of the resonance for the formation of halos have not been simulated.

The Higgs hypothesis. Everything that the earlier paper lists as not established: that a condensate forms in the vacuum, that the rotation group of the extra dimensions is the electroweak gauge group, the scale and the mass of the scalar particle, and the masses of ordinary fermions. The exponent n_{EW} has not been computed for the model itself; the statement that it is of order one or larger rests on the scale-invariant case and on the general fact that the scale of the condensate vanishes at the critical coupling.

XIV. CONCLUSIONS

We asked what becomes of a 16-component Dirac field, and of the superfluid of its pairs, when three of the eight coordinates of a spacetime of signature 7+1 deflate exponentially.

The geometry is simple and rigid. The deflating spacetime is homogeneous, it has 14 to 20 symmetries instead of 36, and it is not a solution of Einstein's equations without matter. With three equal rates it is an exact solution of Einstein-Gauss-Bonnet gravity without matter, for one relation between the rate and the two constants of that theory.

The Dirac field separates into two parts. Fields that do not depend on the deflating coordinates obey, after a change of normalization, exactly the flat equations. Fields that do depend on them are populated by the deflation with a Fermi-Dirac spectrum at the temperature of the rate divided by 2π, and their quanta then gain energy without bound.

The superfluid of hypothesis H1 exists on the deflating spacetime at every moment, with the couplings of that moment, and it acquires a history. The masses do not change; the coupling grows with the inverse volume of the deflating coordinates; the gap grows much faster than the coupling, in weak coupling as the exponential of an exponential; two fermions become bound at a scattering resonance; matter loses its fermion mass and becomes self-bound; and the paired state can follow the deflation only once its gap exceeds a few times the rate divided by the pairing strength. The direction of the order parameter is not selected by the deflation.

For hypotheses H2 and H3 the deflation brings no relief. The rotations that gave the Higgs hypothesis its one structural support on the flat 7+1 spacetime are no longer symmetries of the spacetime. If the superfluid were the Higgs field, the electroweak scale would follow the growing coupling, and atomic clocks would then limit the rate of deflation to about a millionth of the Hubble rate. If gravity is the Einstein theory in eight dimensions, lunar laser ranging limits it to three thousandths of the Hubble rate, and the Hubble constant inferred from the microwave background moves away from the local measurement. In either case nothing of the history just described can have happened within the age of the universe at a constant rate.

These results are quantitative, and they come with the programs that produced them. They mark what a successful theory of this kind would have to supply: a source or a theory of gravity that sustains the deflation, a rate that was large early and is small today, and a reason why the constants of ordinary matter do not follow the volume of the deflating coordinates.

DATA AVAILABILITY

The source code of all programs, the data files from which every figure and every computed number of this paper is generated, the instructions for compiling and running the programs, the supplements that derive each result step by step, and the documents that record the provenance of each result are openly available [4].

Appendix A: Symbols

Tables XIII and XIV list the symbols of this paper with the place of their definition. Symbols of the earlier paper [2] keep their meanings. The complete registry, with the machine checks that no symbol has two meanings, is part of Ref. [4].

TABLE XIII. The symbols of this paper: geometry and the Dirac field.

symbol	meaning	defined in
$x0, \ldots, x7$	the eight coordinates: $x0$ hidden space-like coordinate; $x1, x2, x3$ ordinary space; $x4$ ordinary time; $x5, x6, x7$ the deflating coordinates, space-like	Sec. I
xt; xj; a	a deflating coordinate, $t \in \{5, 6, 7\}$; a flat coordinate other than the time, $j \in \{0, 1, 2, 3\}$; index of the four extra axes, $a \in \{0, 5, 6, 7\}$	Sec. I
g55, g66, g77; rate_t	the three rates of deflation (upright, without subscript; not components of the metric); the rate of xt	Eq. (1)
const5, const6, const7; const_t	the three positive constants of the metric: the scale factors at $x4 = 0$	Eq. (1)
gsum, gsq, gprod	the sum of the rates, the sum of their squares, the sum of their products in pairs	Sec. II
b_t	scale factor of the deflating coordinate xt	Eq. (2)
η; $g_{\mu\nu}$; ${e_A}^\mu$	flat metric; metric of the deflating spacetime; orthonormal frame	Eq. (3)
V_d	proper volume of a unit coordinate cell of the deflating coordinates	Eq. (4)
${\Gamma^\rho}_{\mu\nu}$; ${R^\rho}_{\sigma\mu\nu}$, ${R^\mu}_\nu$, R	Christoffel symbols; Riemann tensor [in Eq. (6) its components in the orthonormal frame], Ricci tensor, curvature scalar	Sec. II B
${\mathcal{G}^\mu}_\nu$; ${\mathcal{H}^\mu}_\nu$; Λ_8, α_{GB}	Einstein tensor; Lanczos tensor of the Gauss-Bonnet term; cosmological constant and coupling of Einstein-Gauss-Bonnet gravity	Sec. II C
$\mathcal{X}_{\text{scale}}$, $\mathcal{X}_{(s)}$	the scaling Killing vector; the three special Killing vectors of the case of equal rates	Eqs. (11), (B8)
K_{AB}; $\bar{x}4$, $\bar{x}t$	sectional curvature of the plane of the frame directions A and B; coordinates adapted to the case of equal rates	Appendix B
P_μ; ς	conserved momenta of a classical particle; affine parameter	Sec. II E
γ^A, γ^μ; Ω_μ, $\omega_{\mu AB}$	Dirac matrices of the frame and in coordinate components; spin connection and its coefficients	Sec. III
χ, $P^{L,R}$; C^-	chirality matrix and the two chiral projectors; the matrix that turns each Dirac matrix into minus its transpose	Secs. V, VI
ψ_{1671d}; $\hat{\psi}$	the 16-component field on the deflating spacetime; the weighted field $\sqrt{V_d}\,\psi$	Eqs. (16), (18)
h, $\mathsf{h}_\perp$; β, α^A, α^d; $\bar{\psi}$	Hamiltonian of a mode, and its part without the scale factors; γ^4, $\gamma^4\gamma^A$, and the matrix α along the wave vector in the deflating coordinates; the adjoint $\psi^\dagger\beta$	Eqs. (19), (26)
p_j, k_t; $E_\perp$; k_d; u_d	wavenumbers along the flat and the deflating coordinates; $(m^2 + \sum_j p_j^2)^{1/2}$; physical wavenumber along the deflating coordinates; k_d/rate_t	Eqs. (20), (22)
w; Z_+, Z_-	one of the eight constant spinors from which the exact solutions are built; two cylinder functions whose orders differ by one	Eq. (23)
n_{cr}; T_{defl}	number of quanta that the deflation creates in a mode, per internal state; temperature of their spectrum	Eqs. (24), (25)

Appendix B: Curvature of the deflating metric

This appendix derives the Einstein tensor and the Lanczos tensor of Sec. II in a form that can be checked by hand, and gives the coordinates and the additional symmetries of the case of equal rates.

1. Curvatures on pairs of directions

In the orthonormal frame the Riemann tensor (6) of the deflating metric is diagonal on pairs of directions: for each pair of frame directions $A \neq B$ it has one independent component, the sectional curvature $K_{AB} = K_{BA}$ of the plane that the two directions span. In the convention in which space-like components of the metric are positive, which this appendix uses, the sectional curvatures are, for both signatures,

$$K_{4t} = \text{rate}_t^2, \qquad K_{ts} = \text{rate}_t\,\text{rate}_s \quad (t \neq s), \tag{B1}$$

and $K_{AB} = 0$ for every pair that contains one of the flat directions $0, 1, 2, 3$. For a Riemann tensor of this kind the mixed components of the Einstein tensor and of the Lanczos tensor are diagonal, and

$${\mathcal{G}^A}_A = -\sum_{\{B,C\}} K_{BC}, \tag{B2}$$

$${\mathcal{H}^A}_A = -4 \sum_{\{B,C\},\{B',C'\}} K_{BC}\,K_{B'C'}, \tag{B3}$$

without summation over A. The first sum runs over the pairs of directions that do not contain the direction A. The second runs over the unordered couples of two such pairs that have no direction in common; a set of four directions contributes its three divisions into two pairs. As a test, for a space of constant curvature in eight dimensions, where all K_{BC} equal one number, the first sum has 21 terms and the second 3×35, so that the two formulas give -21 times that number and -420 times its square, the known values $-\frac{1}{2} \cdot 7 \cdot 6$ and $-\frac{1}{2} \cdot 7 \cdot 6 \cdot 5 \cdot 4$.

TABLE XIV. The symbols of this paper, continued: the model, the superfluid, and the cosmological part.

symbol	meaning	defined in
$\mathcal{L}^{\rm VS}_{1671d}$; $\Phi_{\rm VS}$, $\hat{\Phi}_{\rm VS}$	Lagrangian of the model; the Vorton-Scalar and its weighted form $\sqrt{V_d}\,\Phi_{\rm VS}$	Eq. (26)
$\lambda_{\rm VS}$, $v_{\rm VS}$, $y_{\rm VS}$	self-coupling of the Vorton-Scalar, the constant that fixes its vacuum value, its Yukawa coupling	Sec. V
$U(1)_F$, $U(1)_{\rm VS}$; ϵ	the symmetries of fermion number and of chirality; parameter of a symmetry transformation	Sec. V
m, m_s; $g_{\rm VS}$, $\alpha_{\rm VS}$	fermion mass, mass of the radial mode; coupling of both modes to the fermions, and $g^2_{\rm VS}/4\pi$	Eq. (27)
$D_{\rm VS}$, C; D_c	the two numbers that decide the state of matter; critical value of $D_{\rm VS}$ for a bound pair	Eq. (31)
τ_d; $\tau_{\rm pair}$	the deflation clock; the clock measured from the binding of the pair	Eqs. (30), (40)
$\varphi_{\rm bos}$, $m_{\rm bos}$, ϖ_b	a mode of the Vorton-Scalar, its mass, the frequency of its weighted form	Sec. V D
F_S, F_P; $W(k,k')$	exchange amplitudes of the radial and of the phase mode; kernel of the gap equation (always written with its two arguments)	Sec. VI
λ, λ_S, λ_P; Δ_F; T_c	pairing strength, and its parts from the radial and from the phase mode; gap at the Fermi momentum; critical temperature	Sec. VI
k_F, E_F, v_F, $N(0)$	Fermi momentum, Fermi kinetic energy, Fermi velocity, density of states at the Fermi surface per internal state	Secs. VI, IX
$a_{\rm sc}$, E_b	scattering length of two fermions; binding energy of the pair	Sec. VII
Ξ_d; $\Delta_{\rm eq}$; $E_{\rm sh}$; $n_{\rm ex}$	the parameter that decides whether the paired state follows the deflation; equilibrium gap; half-width of the shell of states; density of broken pairs	Sec. VIII
ξ_k, E_k, κ, n_k	single-particle energy, quasi-particle energy, pair correlation and occupation of a pair of states	Eq. (43)
$V[\Delta]$; ϕ; m^*; $T_{1/2}$	effective potential of the paired state; expectation value of the Vorton-Scalar; fermion mass in matter; temperature at which the fermions have lost half of their mass	Sec. IX
Δ^a, Δ_t; M; $d_\pm$	order parameter (frame components), and its coordinate components along the deflating coordinates; gap matrix; the two fourfold gaps of a state	Sec. X
$SU(2)_A$, $SU(2)_B$, $U(1)_E$	the factors of the internal symmetry of the fields that do not depend on the extra coordinates	Sec. X C
P, P_c; w_z; θ_{12}	generator of $U(1)_E$ and the product of the five matrices of the sector; spinor of a zero mode; angle around a vortex line	Sec. X D
$a_{\rm sp}$, H, H_0; G_N, G_8	scale factor of ordinary space, Hubble rate, Hubble constant; Newton's constant and the constant of gravitation in eight dimensions	Sec. XI
ρ, ρ_m; ω_b, ω_c	energy density in ordinary space, and that of matter; the densities of baryons and of cold dark matter of the standard model of cosmology	Sec. XI
θ_*, r_s; t_0	angle under which the sound horizon at last scattering is seen, and that sound horizon; age of the universe	Sec. XI
δ_m, f_g, σ_8	density contrast of matter, its growth rate, amplitude of clustering	Sec. XI
$v_{\rm EW}$, $n_{\rm EW}$; α_c, ϱ_c	electroweak scale and the exponent with which it would follow the coupling under H2; critical coupling of the vacuum condensate, and an abbreviation	Sec. XII
m_p/m_e; $G_{\rm Fermi}$; $\alpha_{\rm em}$	ratio of the proton mass to the electron mass; Fermi constant; fine-structure constant	Sec. XI
H1, H2, H3; A1, A2; HV	the three hypotheses about the superfluid; the two assumptions of the cosmological part; the assumption that the vacuum value of the Vorton-Scalar sets the electroweak scale	Secs. I, XI

The Einstein tensor. With Eq. (B1), Eq. (B2) gives

$$\mathcal{G}^4{}_4 = -\mathrm{gprod}, \qquad \mathcal{G}^j{}_j = -(\mathrm{gsq} + \mathrm{gprod}),$$
$$\mathcal{G}^5{}_5 = -(\mathrm{g66}^2 + \mathrm{g77}^2 + \mathrm{g66\,g77}), \tag{B4}$$

and the corresponding expressions for $\mathcal{G}^6{}_6$ and $\mathcal{G}^7{}_7$. Because $\mathrm{gsum}^2 = \mathrm{gsq} + 2\,\mathrm{gprod}$, these are Eq. (7) multiplied by $-\eta_{44}$, the factor that relates the two conventions.

The Lanczos tensor. A couple of pairs contributes to Eq. (B3) only if neither pair contains a flat direction, that is, only if the four directions are $4, 5, 6, 7$. For a flat direction $A = j$ these are allowed, and their three divisions into two pairs give

$$K_{45}K_{67} + K_{46}K_{57} + K_{47}K_{56} = \mathrm{g55\,g66\,g77\,gsum}, \tag{B5}$$

which is Eq. (9). For $A = 4, 5, 6, 7$ the direction A is itself among the four, and these components vanish. The Lovelock tensor of the third order is built in the same way from sets of six directions, and every such set contains a flat direction: it vanishes.

The field equations. In Eq. (8) the components along $x4$ and along the deflating coordinates read $\mathcal{G}^4{}_4 + \Lambda_8 = 0$ and $\mathcal{G}^t{}_t + \Lambda_8 = 0$. Since $\mathcal{G}^5{}_5 - \mathcal{G}^6{}_6 = (\mathrm{g55} - \mathrm{g66})\,\mathrm{gsum}$,

they can hold together only if the three rates are equal, and then $\Lambda_8 = 3\,\mathrm{g55}^2$. The components along the flat coordinates read $-6\,\mathrm{g55}^2 + \Lambda_8 - 12\,\alpha_{\mathrm{GB}}\,\mathrm{g55}^4 = 0$, which gives $\alpha_{\mathrm{GB}} = -1/(4\,\mathrm{g55}^2)$: Eq. (10). The program verifies Eqs. (B2) and (B3) for the deflating metric in exact arithmetic, and an independent program obtains the same Lanczos tensor from its standard definition by computer algebra, for arbitrary rates and constants.

2. Equal rates: a space of constant curvature

Let $\mathrm{g55} = \mathrm{g66} = \mathrm{g77}$, and introduce in place of $x4, \ldots, x7$ the coordinates

$$\bar{x}4 = \frac{e^{\mathrm{g55}\,x4}}{\mathrm{g55}}, \qquad \bar{x}t = \mathrm{const}_t\, xt \quad (t = 5, 6, 7). \tag{B6}$$

The part of the line element that contains $x4, \ldots, x7$ becomes

$$\eta_{44}\, dx4^2 - e^{-2\,\mathrm{g55}\,x4} \sum_t d\bar{x}t^2 = \frac{\eta_{44}\, d\bar{x}4^2 - \sum_t d\bar{x}t^2}{(\mathrm{g55}\,\bar{x}4)^2}. \tag{B7}$$

With $\eta_{44} = +1$ this is the line element of de Sitter space of the radius $1/\mathrm{g55}$, in the half of its flat slicing in which space contracts. De Sitter space is familiar as the geometry of inflation, where the same line element is used with the opposite direction of time. The line element (B7) is a multiple of a flat one. The special conformal transformations of that flat metric along the three axes $\bar{x}5, \bar{x}6, \bar{x}7$ leave the factor $1/\bar{x}4^2$ unchanged and are therefore symmetries. Their generators are, for $s = 5, 6, 7$,

$$\begin{aligned} \mathcal{X}_{(s)} = {} & 2\,\bar{x}s \left[\bar{x}4\,\frac{\partial}{\partial \bar{x}4} + \sum_t \bar{x}t\,\frac{\partial}{\partial \bar{x}t}\right] \\ & - \Big(\sum_t \bar{x}t^2 - \bar{x}4^2\Big)\frac{\partial}{\partial \bar{x}s}, \end{aligned} \tag{B8}$$

with $\bar{x}4\,\partial/\partial\bar{x}4 = \mathrm{g55}^{-1}\partial_4$. Together with the 14 Killing vectors of the general case and the three rotations of $x5, x6, x7$ they are the 20 Killing vectors of the case of equal rates.

Appendix C: Machine checks, and how the results are reproduced

All physics of this paper is computed by one program, written in Rust, which uses the program of the earlier paper as a library and no other library. Each of its nine commands writes data files and a list of checks; a check marked "exact" is an identity verified in exact rational arithmetic, and a check marked "numerical" states its accuracy. Table XV lists them. The central results are verified a second time by a program in Python that shares no code with the first and uses other methods: computer algebra with symbolic rates and constants instead of rational arithmetic, a general-purpose integrator and the Bessel functions of a library instead of the integrator of the program. That second program was itself tested by introducing twenty deliberate errors into it, one at a time; it detected nineteen, and the twentieth, a change of the term gprod of Eq. (51), cannot be detected by a comparison of Hubble constants for the reason given in Sec. XI and is covered by the symbolic check of the Einstein tensor.

No computed number in a table of this paper was typed by hand: the tables are written by a script from the data files of the program. The figures are drawn by a script from the same files. The repository [4] contains the programs with their unit tests, the data, the instructions for compiling and running everything, four supplements and a document on the Dirac matrices of the deflating spacetimes that derive each result of this paper step by step, and the record of the errors that were made and corrected during the work.

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[2] P. L. Nash, 16 component Dirac fermion condensate singlet superfluid on 7+1 spacetime (2026), manuscript, source code and data at https://github.com/once-ere/spacetimes.

[3] P. L. Nash, 16 component Dirac fermion condensate singlet superfluid on 4+4 deflating spacetime (2026), companion paper.

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TABLE XV. The machine checks behind this paper: the commands of the program, what each verifies or computes, and the number of its checks that pass. The checks of both deflating spacetimes are counted, since the program treats the two together.

command	content	checks passed
`matrices`	the Dirac matrices of the frame, verified again	106 of 106
`matrices`	geometry, Killing vectors, Einstein and Lanczos tensors, geodesics; matrices in coordinate components, spin connection, Dirac operators	47 of 47
`modes`	solutions of the Dirac equation mode by mode: creation (7+1) and extinction (4+4)	7 of 7
`symmetry`	surviving rotations and boosts, internal symmetry of the sector, zero modes on a vortex line, reduction of the couplings	18 of 18
`channels`	exchange amplitudes in the pair channels; modes of the Vorton-Scalar	7 of 7
`history`	gap, bound pair, matter, vacuum and temperature along the clock; order of events	8 of 8
`quench`	dynamics of the paired state under a growing coupling	5 of 5
`paired`	the paired state and self-bound matter from the effective action	7 of 7
`cosmology`	inferred Hubble constant, bounds on the rate, the exponent n_{EW}	6 of 6
`matter`	dark-matter estimates, the resonance, growth of structure, drifts, temperature of the created quanta	6 of 6
independent program	geometry, completeness of the Killing vectors, Lanczos tensor, spin connection and Dirac operators by computer algebra; exact solutions; vortex zero modes; Hubble constant	33 of 33
registry of symbols	no symbol and no name in a data file has two meanings	6 of 6

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